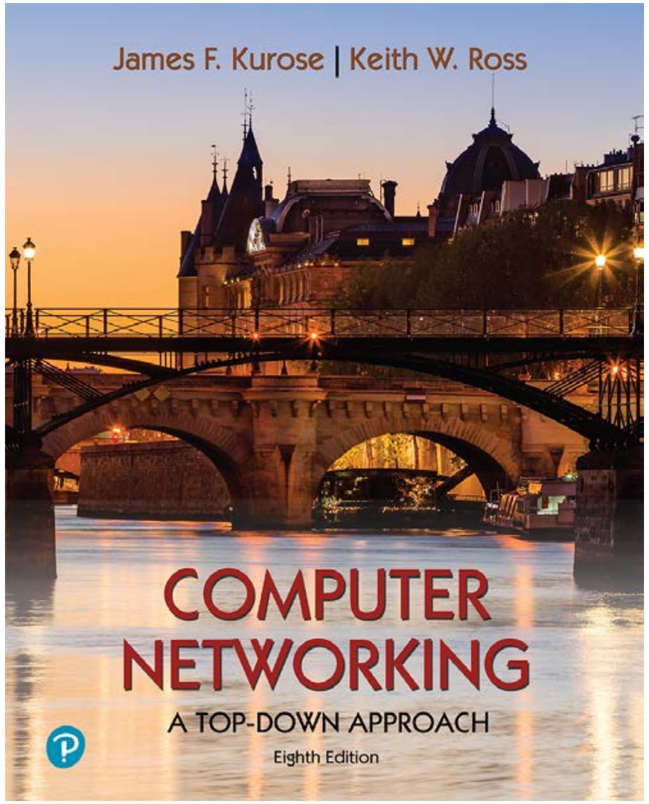
Chapter 3
Transport Layer



Computer Networking: A Top-Down Approach
8th edition
Jim Kurose, Keith Ross
Pearson, 2020

Transport layer: overview

Questions

- how do we get two entities reliably communicate over a channel in which messages can be corrupted or lost
- how two distributed entities synchronize shared state
- how a collection of different entities can adjust their communication rates so that they don't overflow network resources
- Two main transport protocols: UDP, TCP

Transport layer: overview

Our goal:

- understand principles behind transport layer services:
  - multiplexing, demultiplexing
  - reliable data transfer
  - flow control
  - congestion control

Transport layer: overview

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- understand principles behind transport layer services:
  - multiplexing, demultiplexing
  - reliable data transfer
  - flow control
  - congestion control
- learn about Internet transport layer protocols:
  - UDP: connectionless transport
  - TCP: connection-oriented reliable transport
  - TCP congestion control

Transport layer: roadmap

- Transport-layer services
- Multiplexing and demultiplexing
- Connectionless transport: UDP
- Principles of reliable data transfer
- Connection-oriented transport: TCP
- Principles of congestion control
- TCP congestion control
- Evolution of transport-layer functionality

Transport layer: roadmap

- **Transport-layer services**
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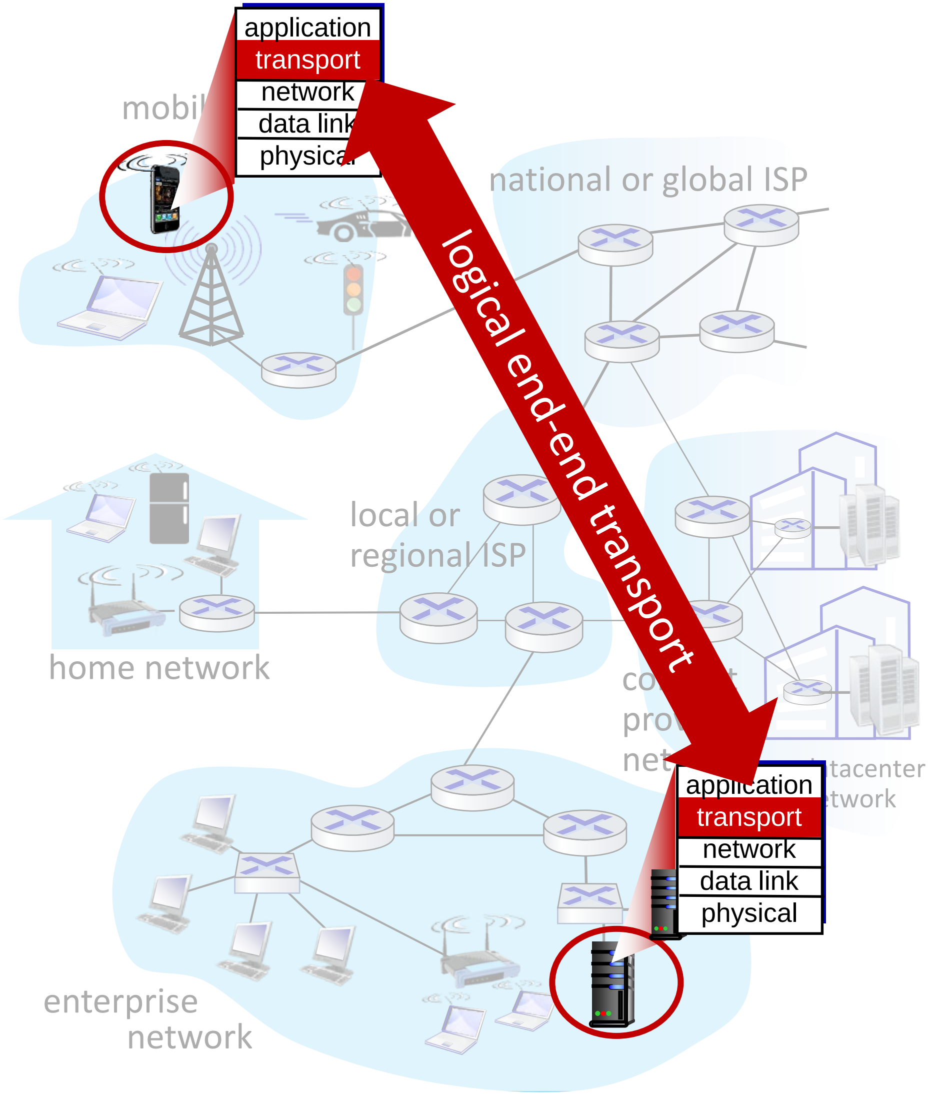
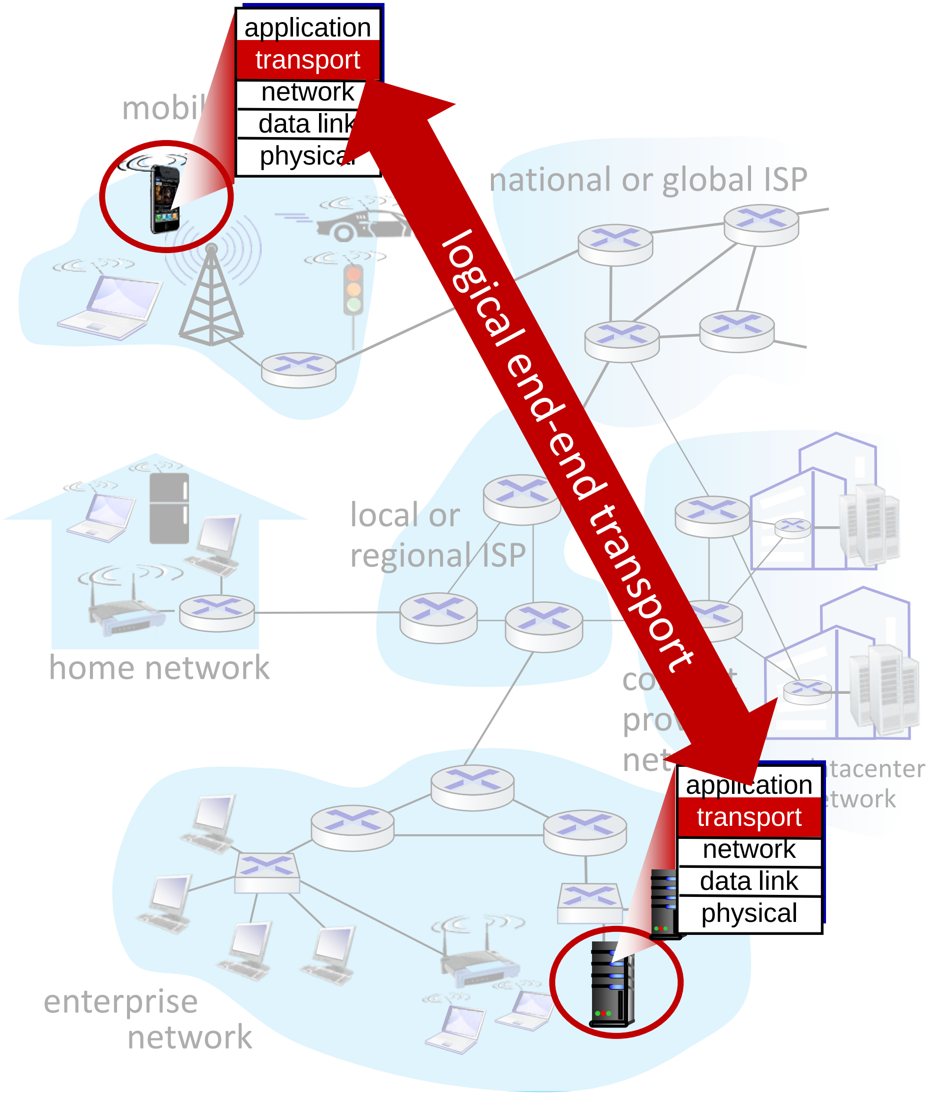
Transport services and protocols

- provide *logical communication* between application processes running on different hosts
- transport protocols actions in end systems:
  - sender: breaks application messages into *segments*, passes to network layer
  - receiver: reassembles segments into messages, passes to application layer
- two transport protocols available to Internet applications
  - TCP, UDP



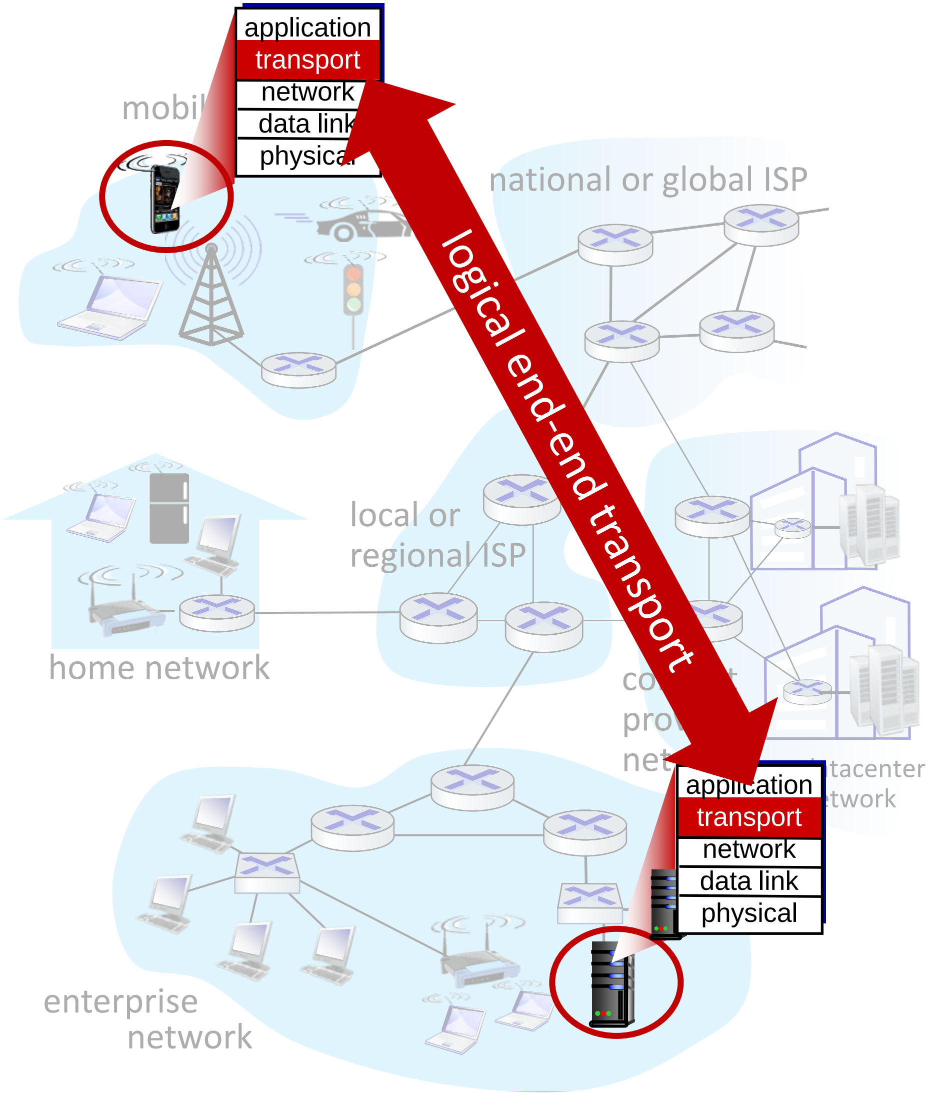
Transport services and protocols

- *Logical communication*
  - *Sender and receiver logically connected to each other by a direct link*
  - *Reality: host may be on different sides of the planet*
  - *Logical perspective: sender and receiver directly connected*

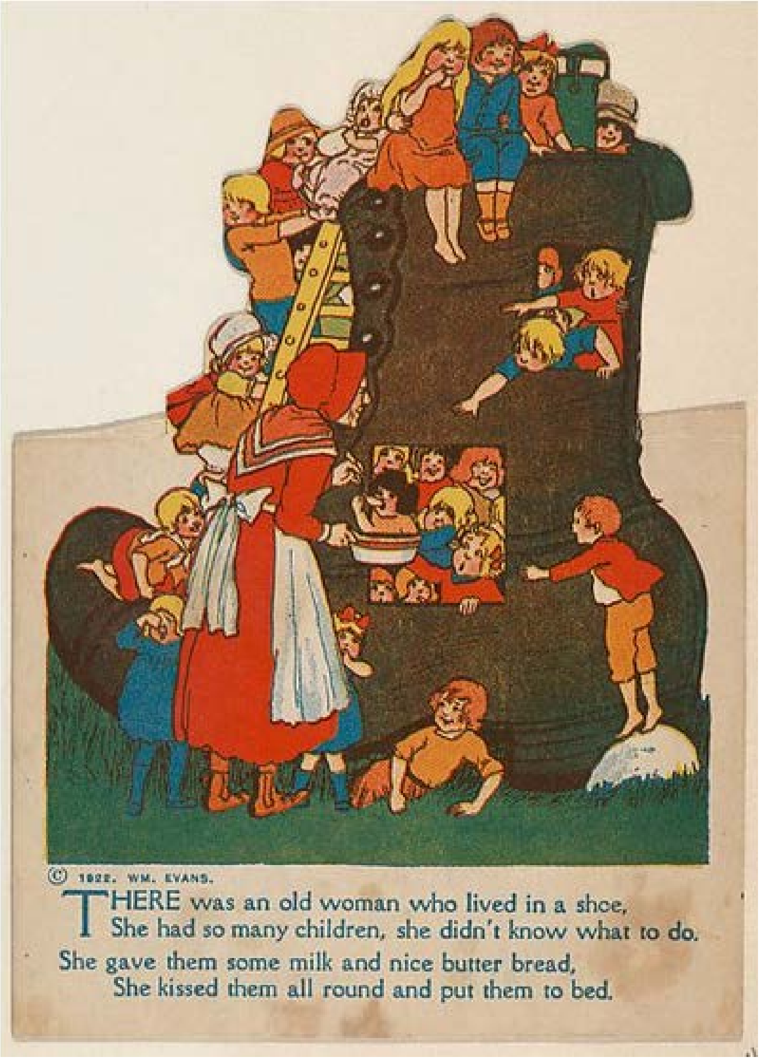


Transport services and protocols

- *Logical communication*
  - *Sender and receiver logically connected to each other by a direct link*
  - *Reality: host may be on different sides of the planet*
  - *Logical perspective: sender and receiver directly connected*
  - *Channel abstraction*
    - *Abstract away what is in between communicating processes*
    - *Focus on:*
      - *Properties of the channel*
      - *How services are implemented given the channel*



Transport vs. network layer services and protocols



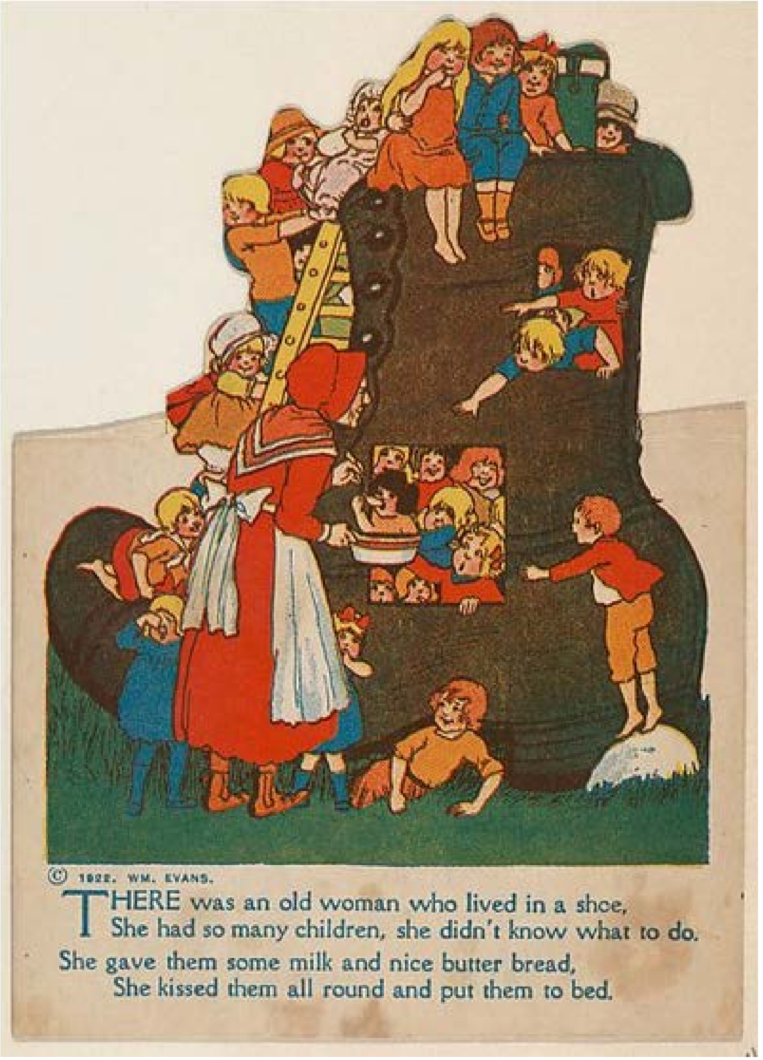
There's a difference between:

"communicating processes"

and

"communicating hosts"

Transport vs. network layer services and protocols



household analogy:

12 kids in Ann's house sending letters to 12 kids in Bill's house:

- hosts = houses
- processes = kids
- app messages = letters in envelopes

Transport vs. network layer services and protocols

- transport layer*:* communication between *processes*
  - relies on, enhances, network layer services

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- network layer: communication between *hosts*

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- network-layer protocol = postal service

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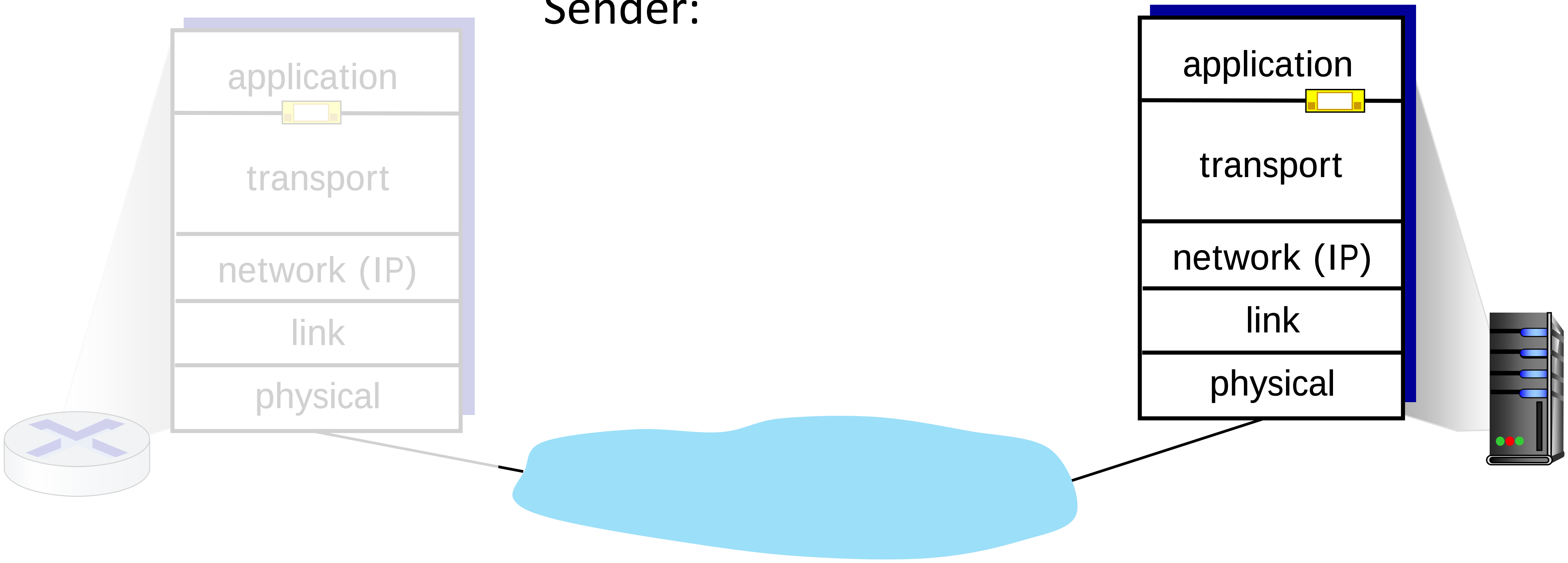
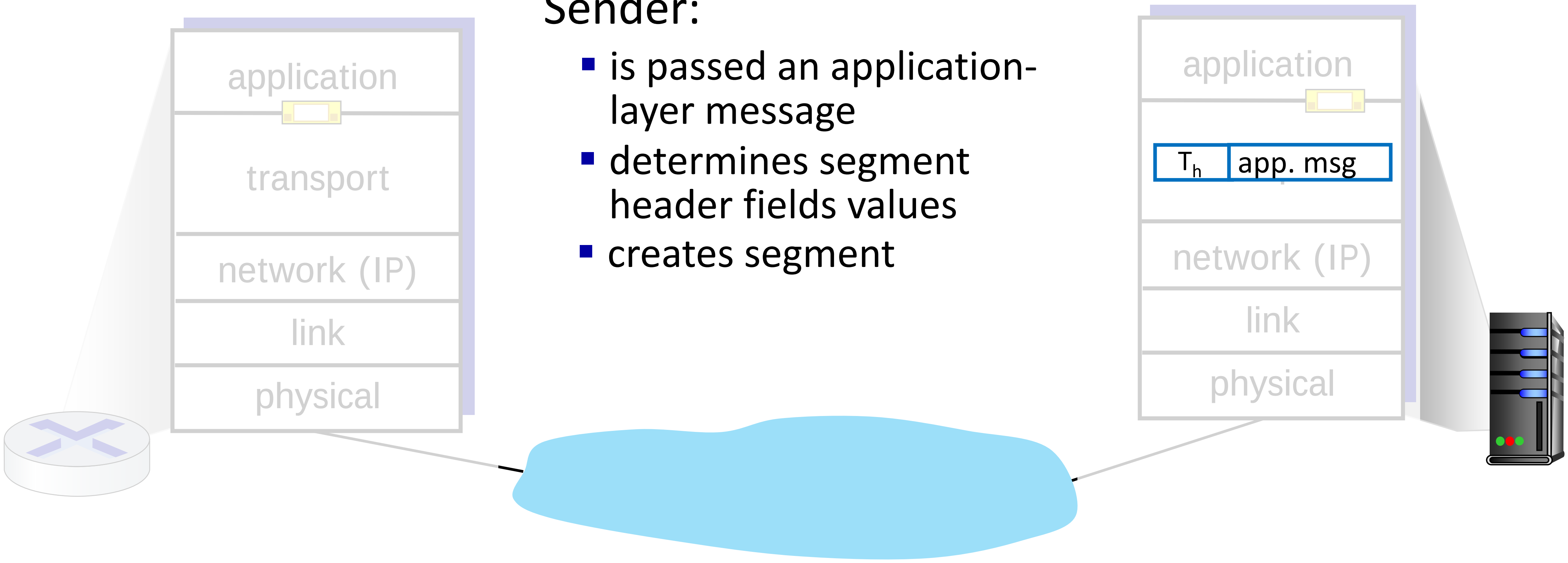
- transport layer*:* what happens inside the house
- network layer: getting messages from one house to another

Transport Layer Actions

Sender:



Transport Layer Actions



Sender:

- is passed an application-layer message
- determines segment header fields values
- creates segment

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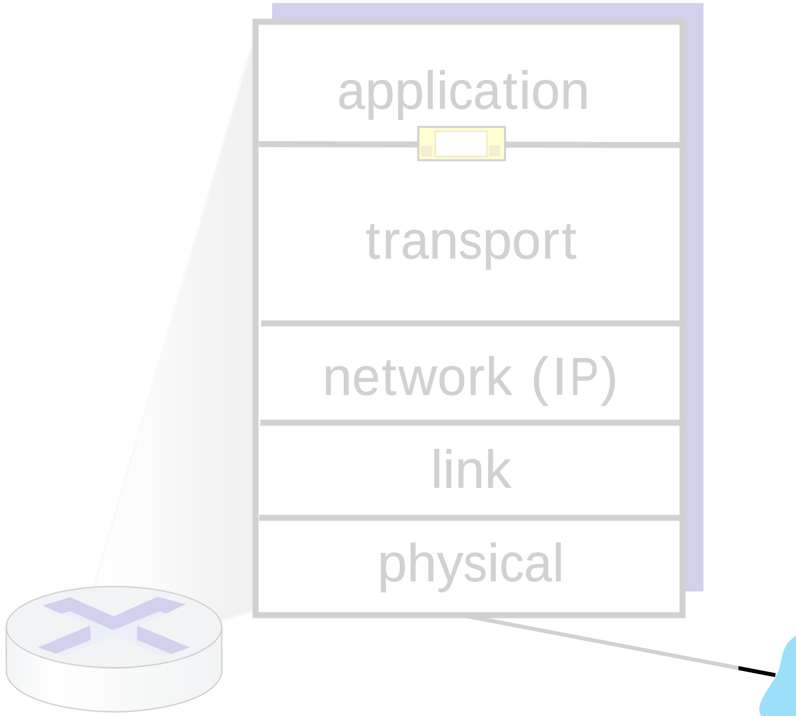

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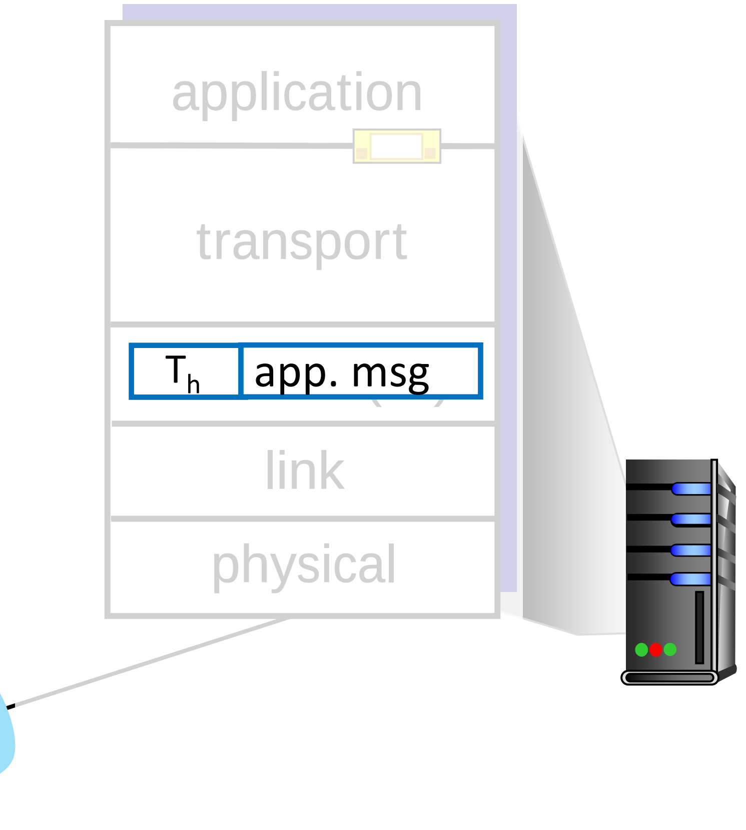

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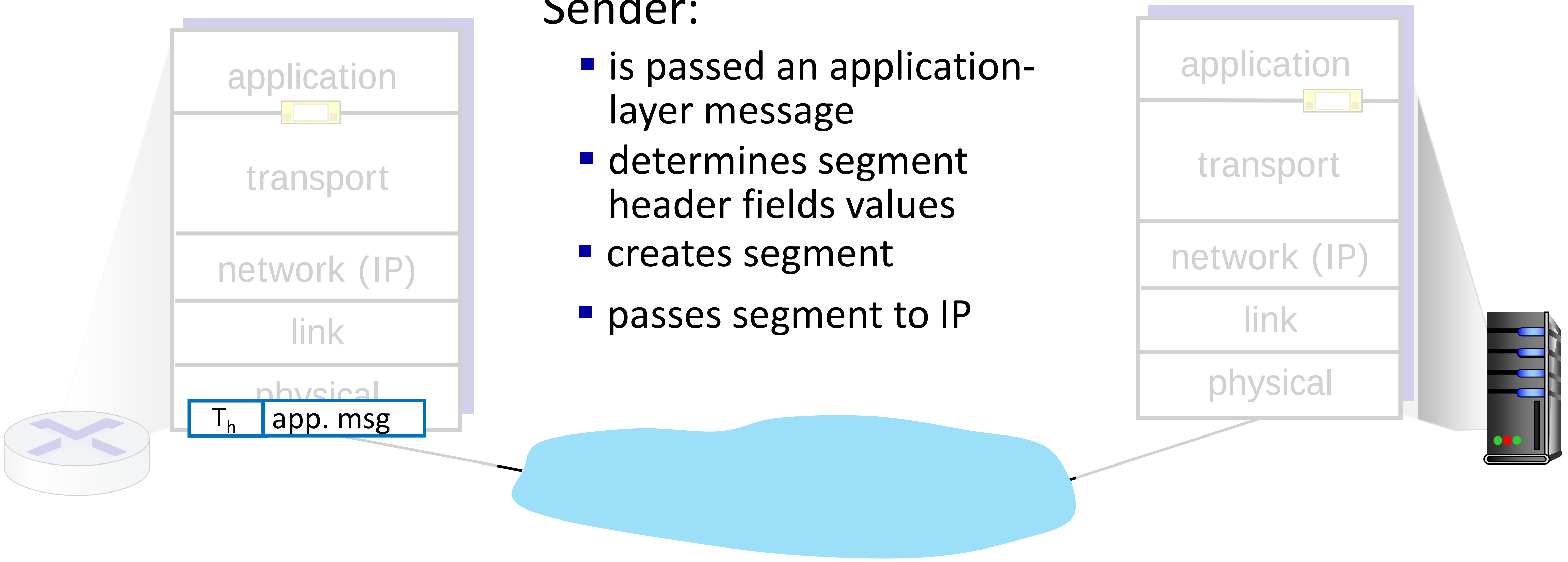
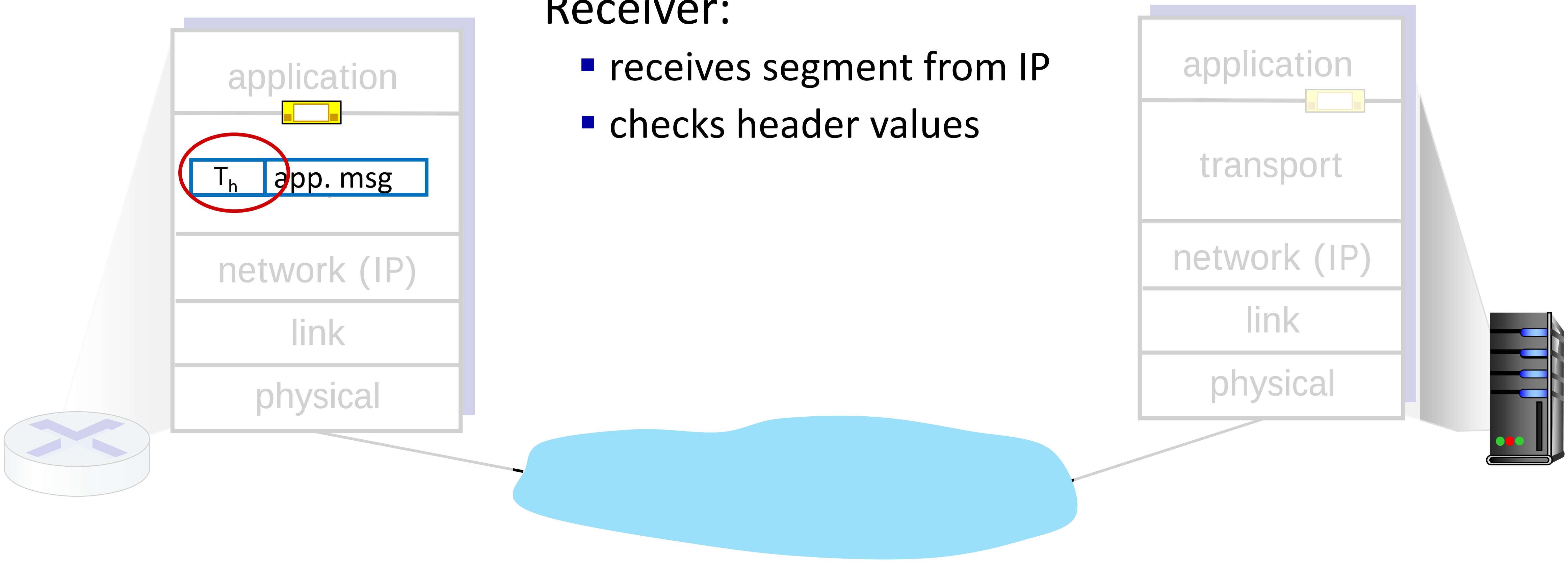


Transport Layer Actions

Receiver:

- receives segment from IP

Transport Layer Actions



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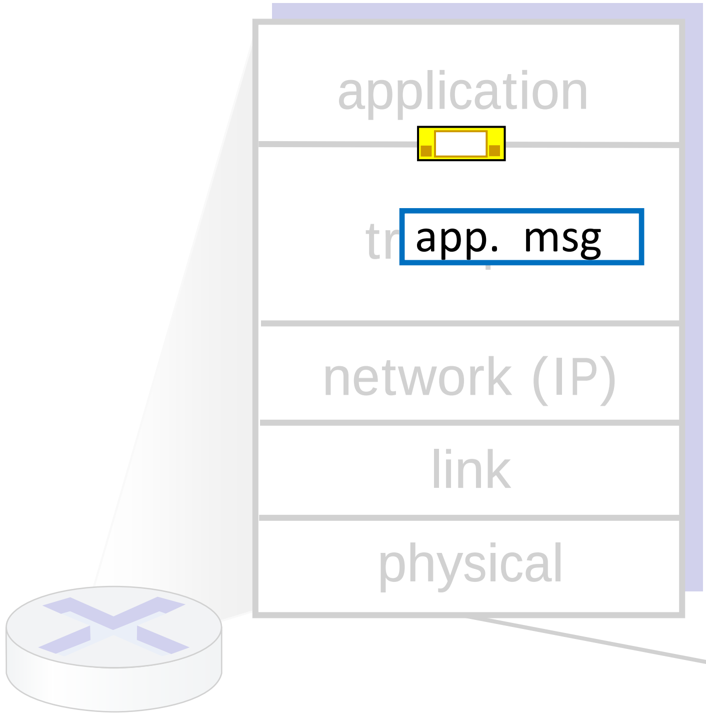

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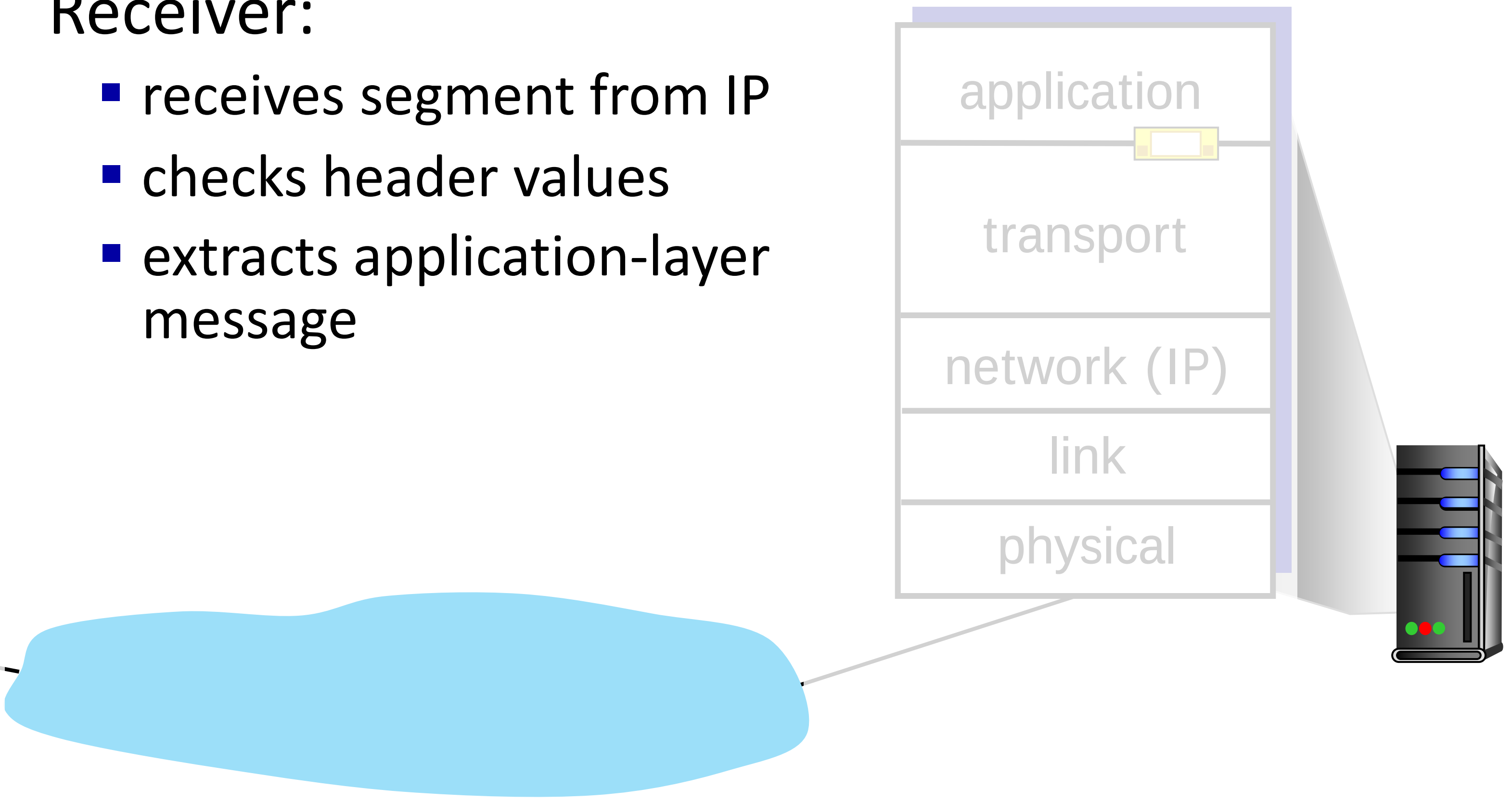

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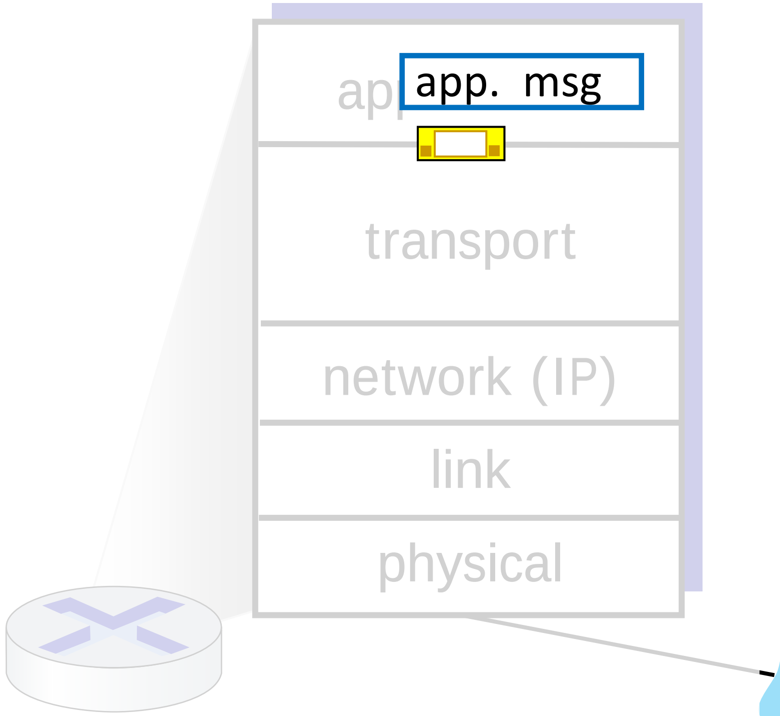
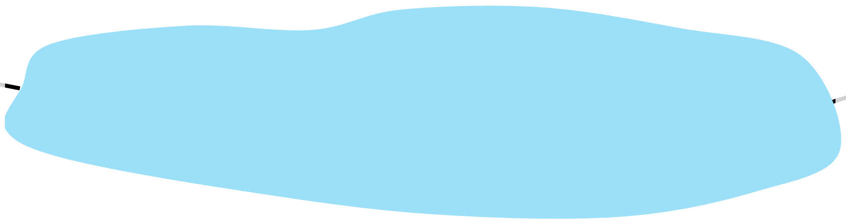
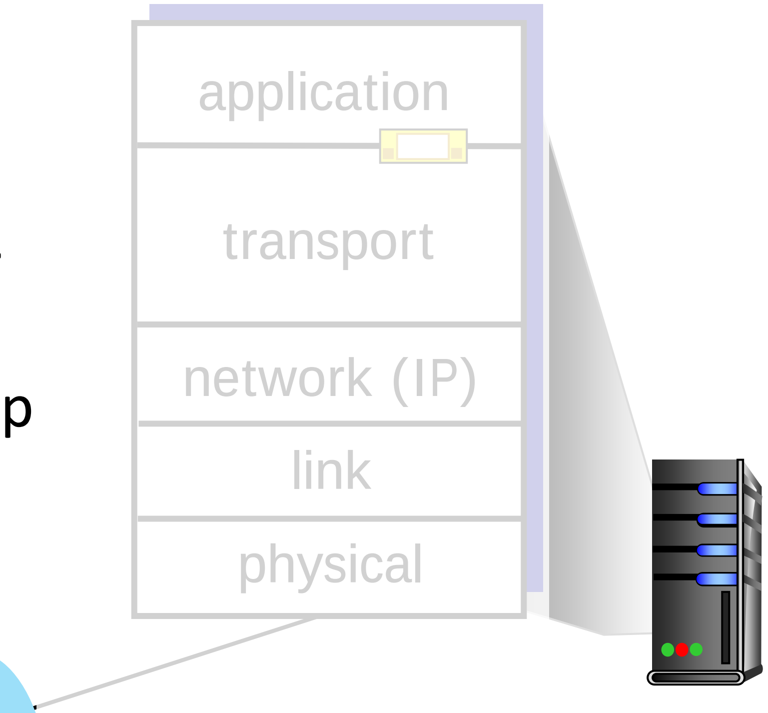


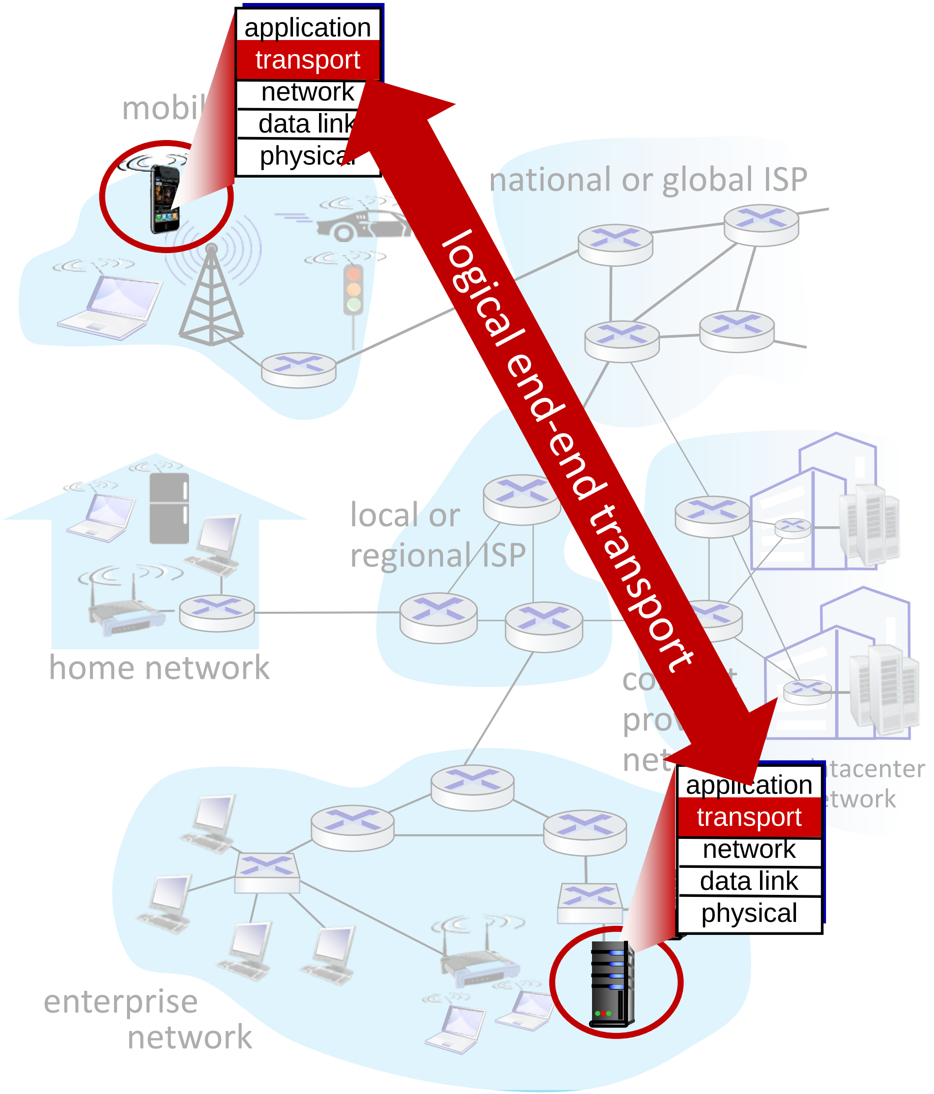
Receiver:

- receives segment from IP
- checks header values
- extracts application-layer message
- demultiplexes message up to application via socket





Two principal Internet transport protocols



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- **TCP:** Transmission Control Protocol
  - reliable, in-order delivery
  - congestion control
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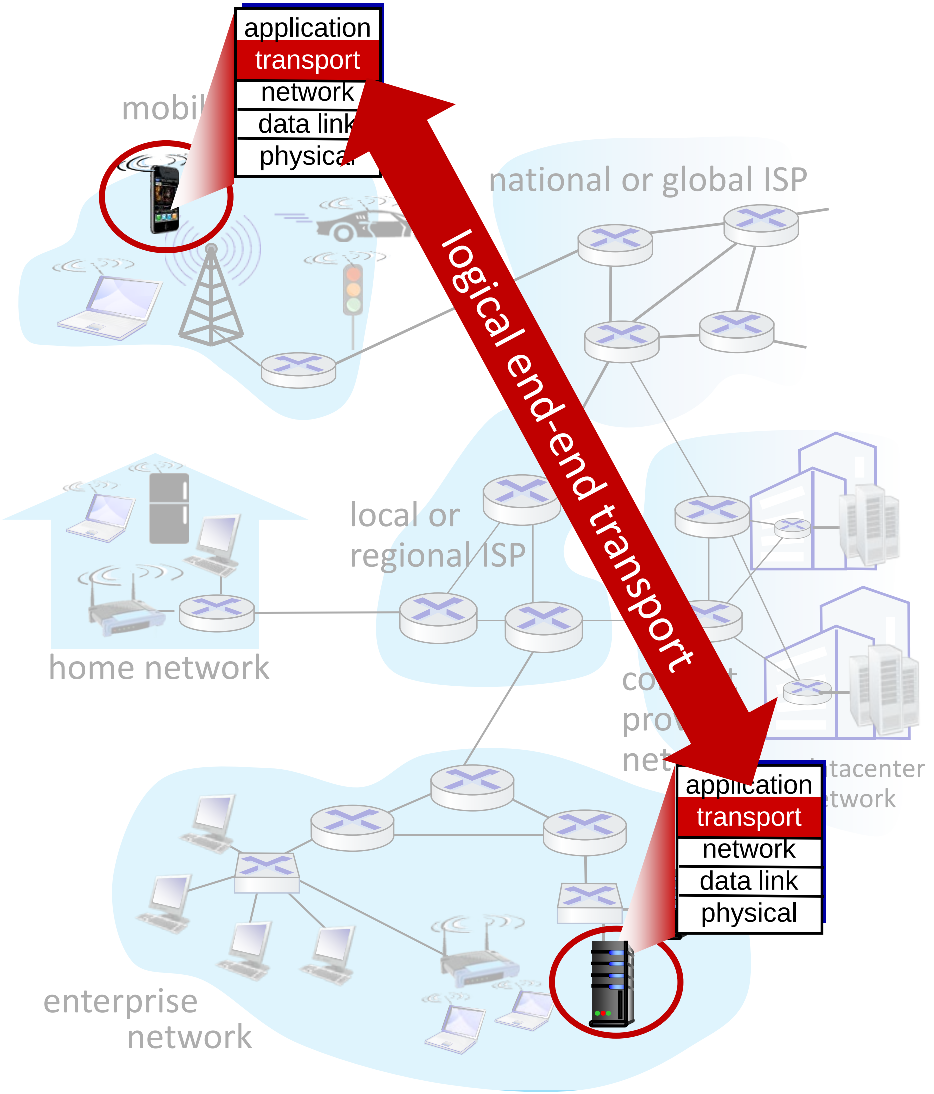

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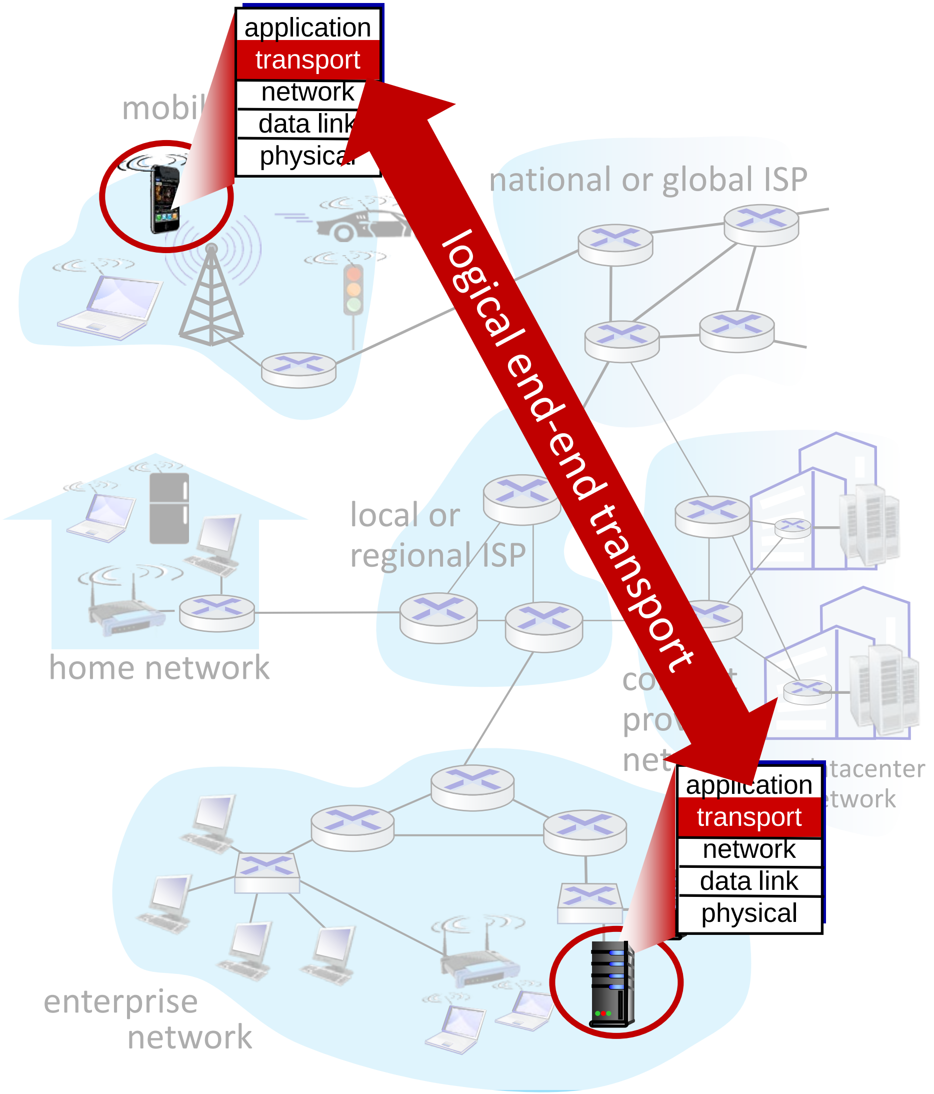
  - no-frills extension of “best-effort” IP



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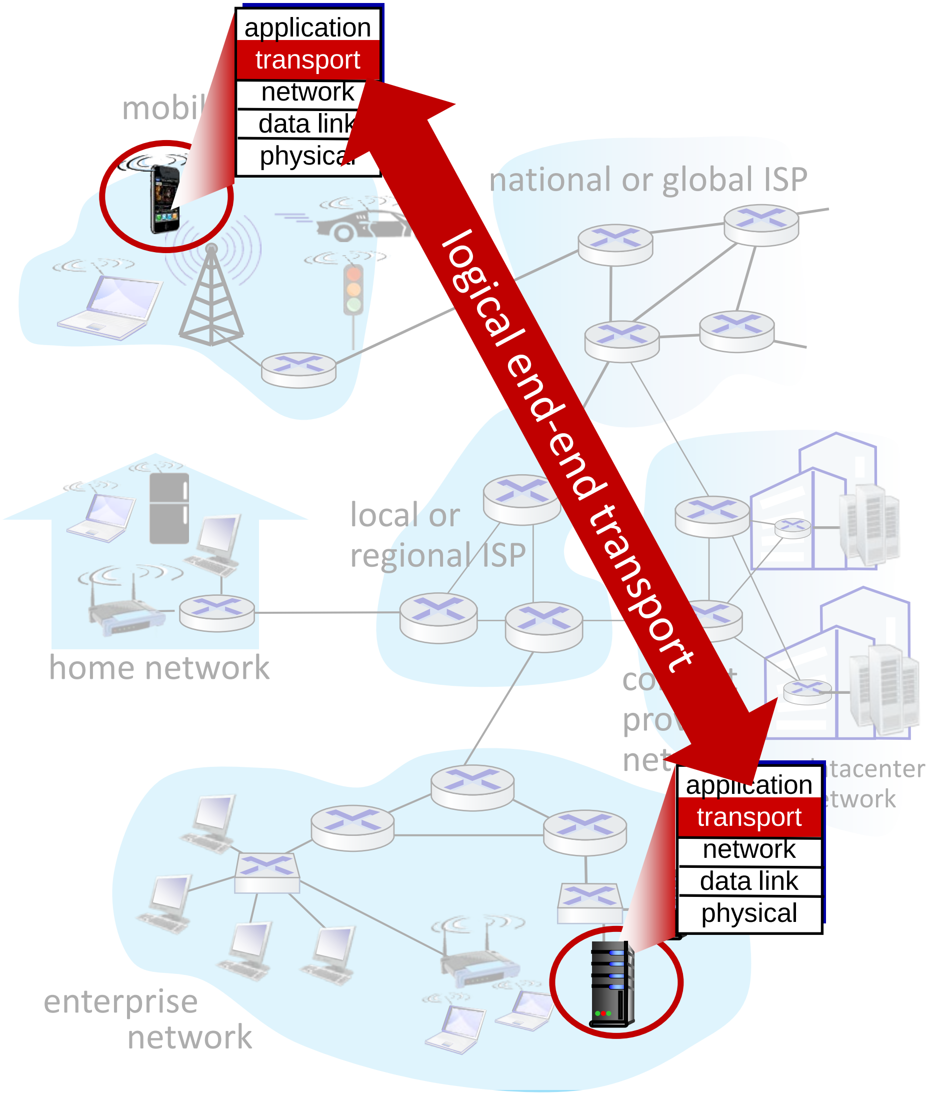

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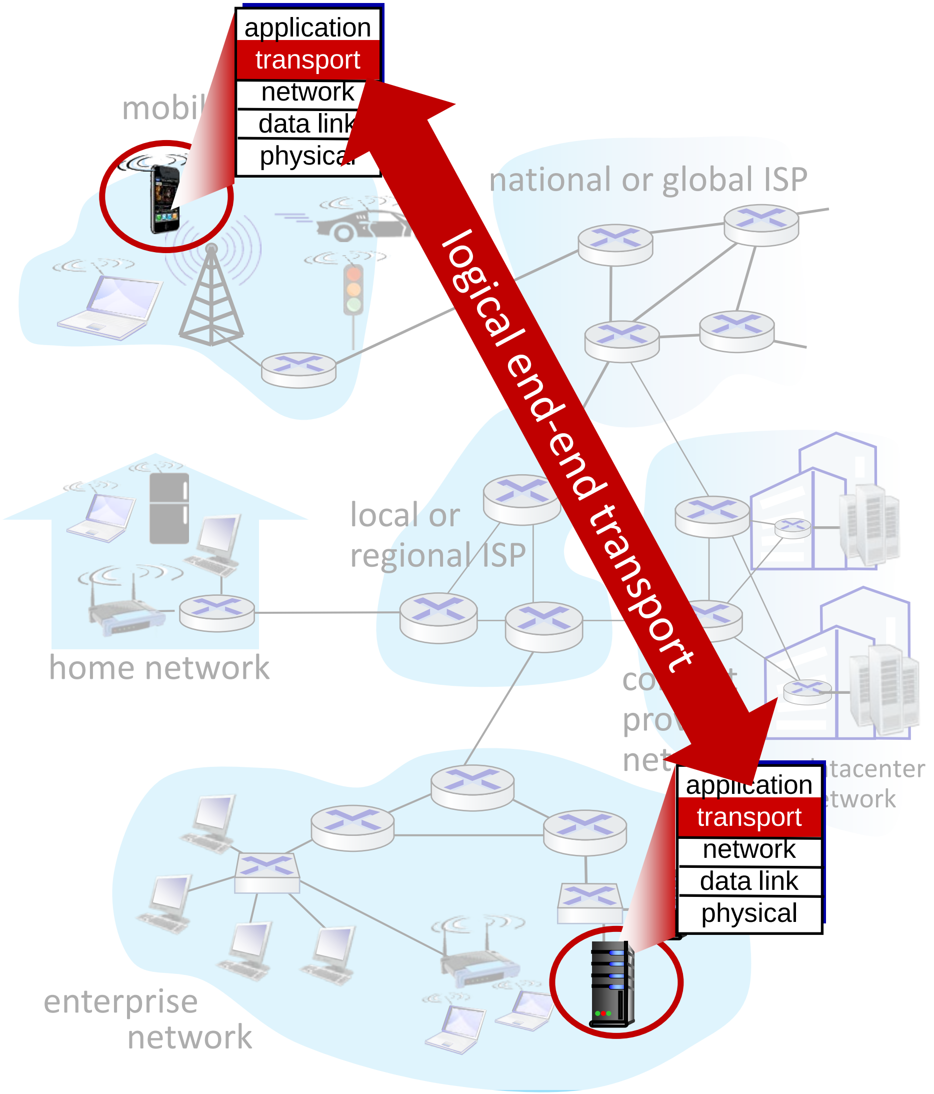
  - bandwidth guarantees



Chapter 3: roadmap

- Transport-layer services
- Multiplexing and demultiplexing
- Connectionless transport: UDP
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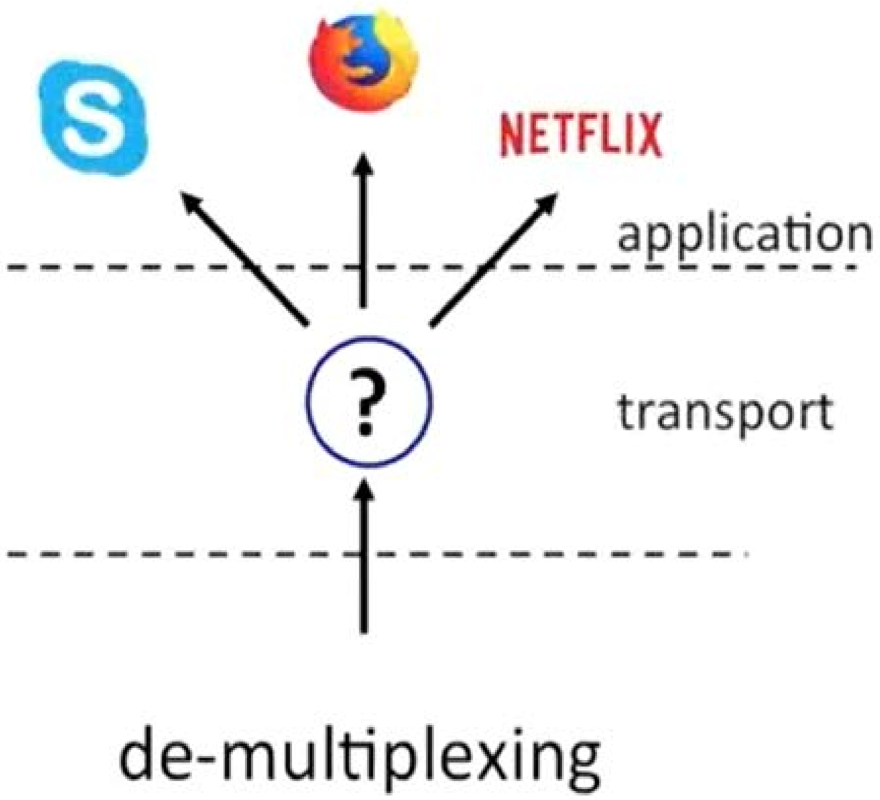
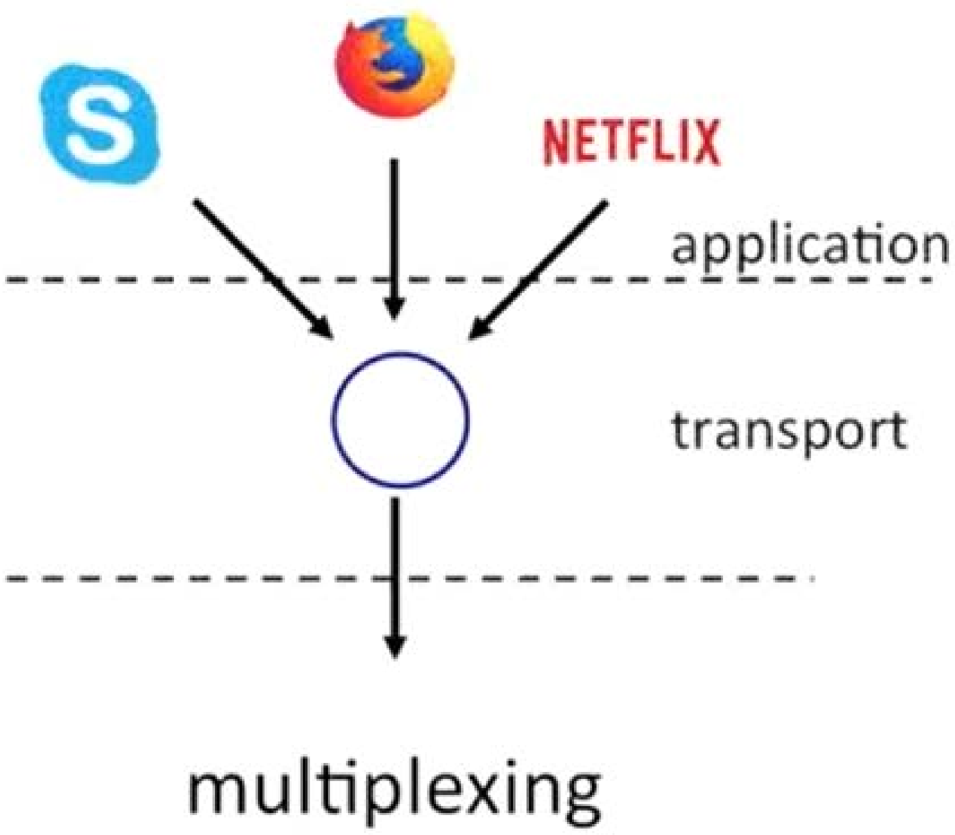
De-multiplexing



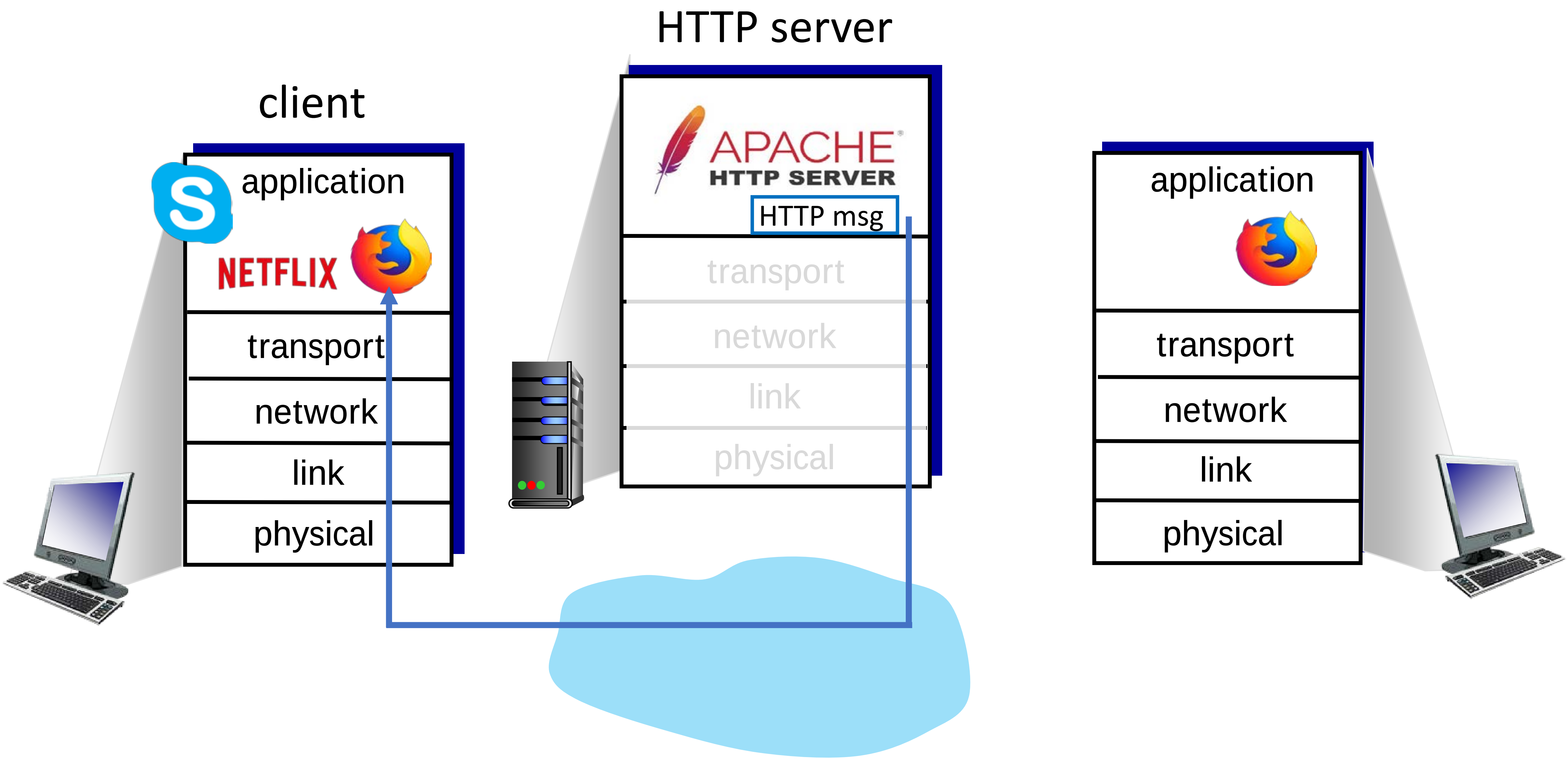
Direct payloads to the appropriate application or protocol

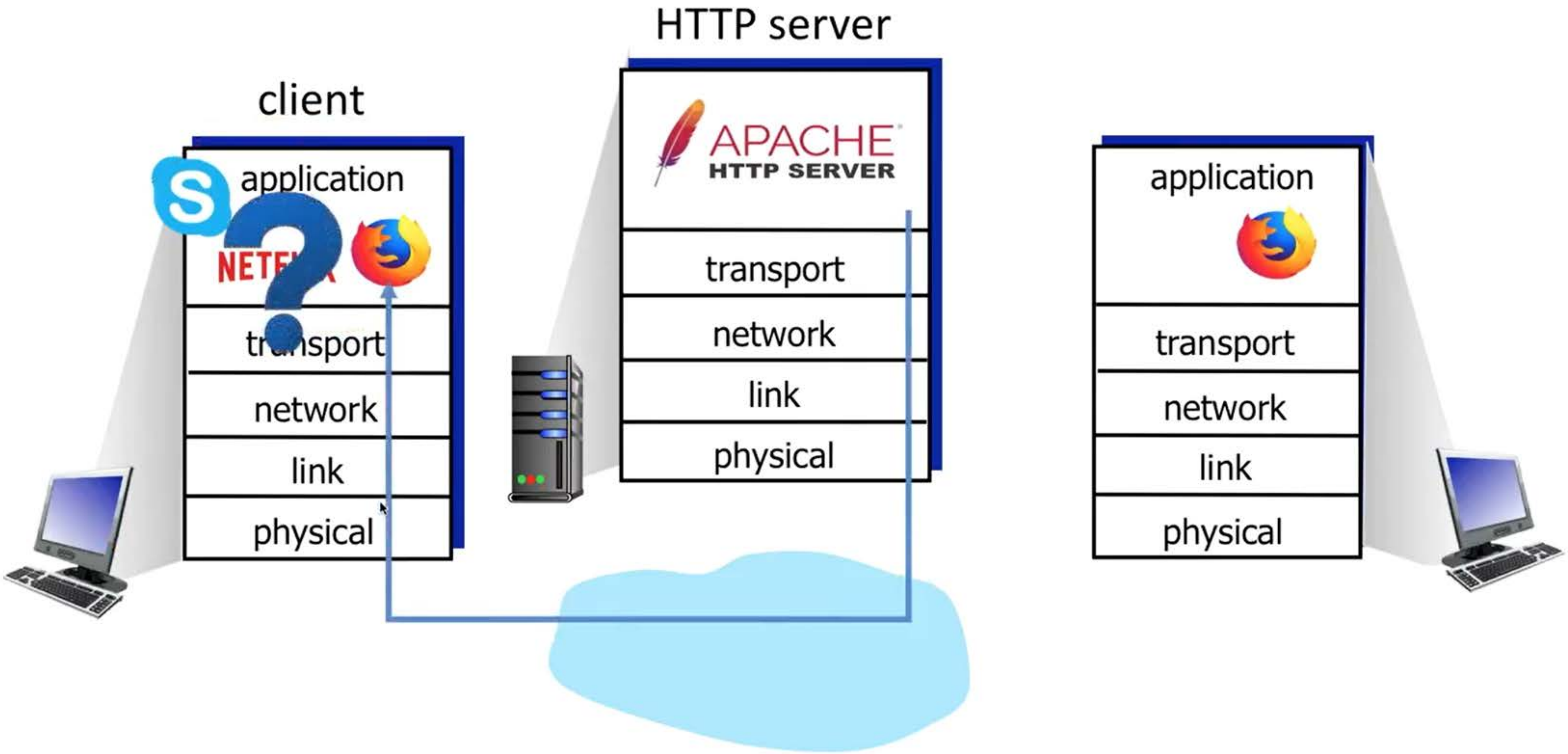
- **Datagram**
  - **Header**
    - **Contains various info i.e. addresses, protocol type, etc**
  - **Payload**
    - **Contains actual data**
    - **Bound for different applications or to different protocols**

Multiplexing



Take application messages sent through multiple sockets, and funnelling them down into IP


HTTP server
client
application
NETFLIX
transport
network
link
physical
APACHE
HTTP SERVER
HTTP msg
transport
network
link
physical
application
transport
network
link
physical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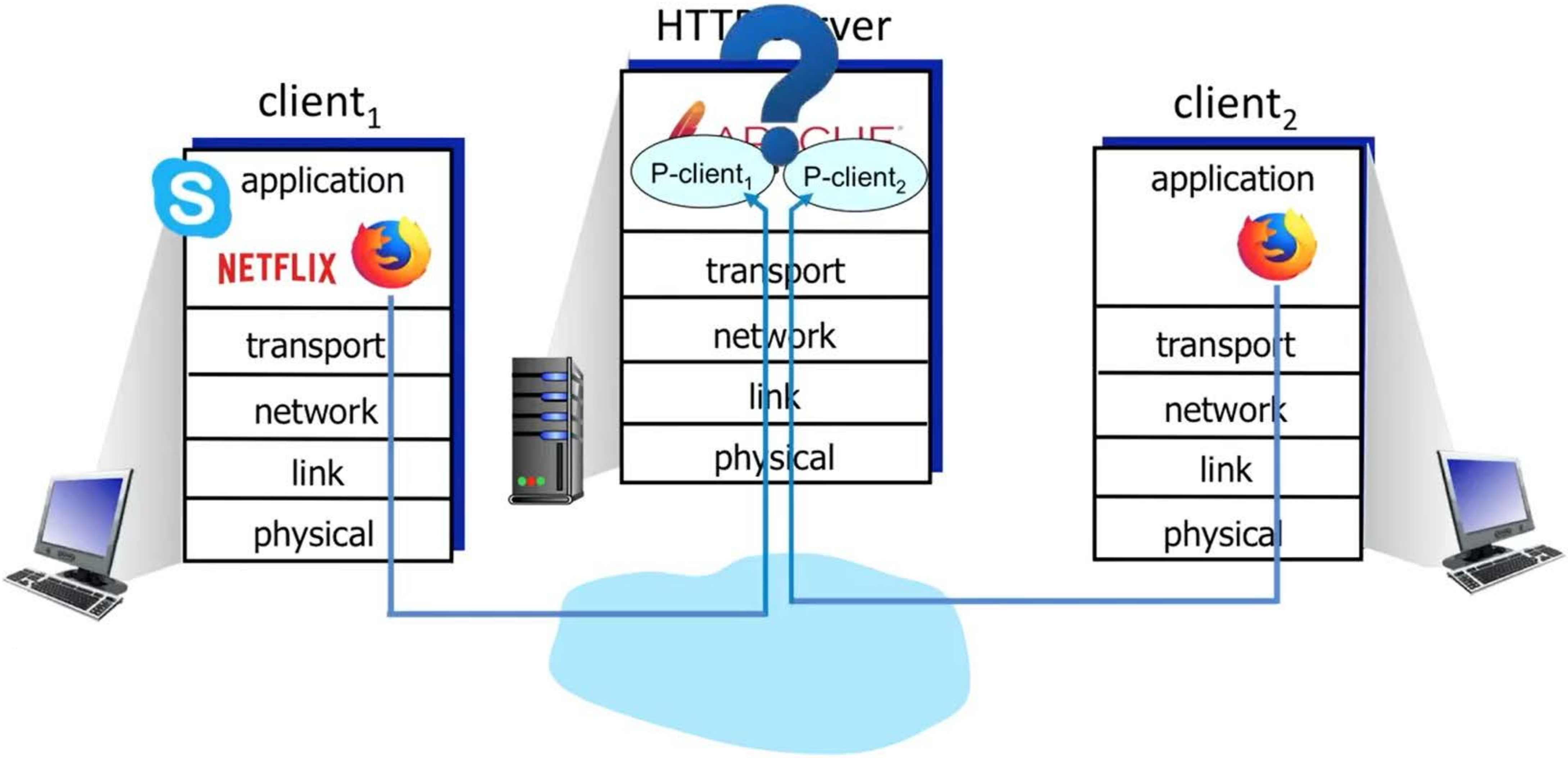
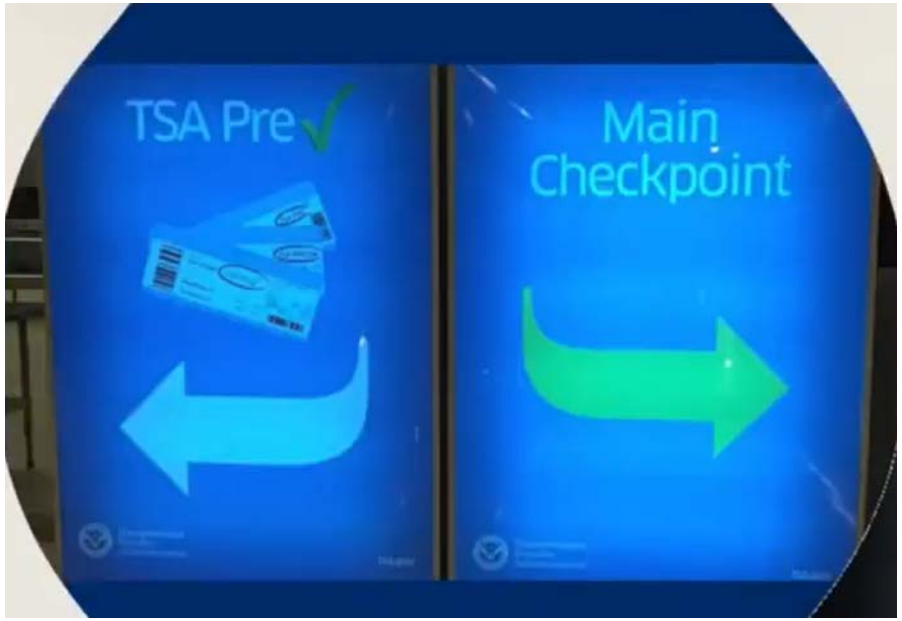


client$_1$
application
NETFLIX
transport
network
link
physical
HTTP server
P-client$_1$
P-client$_2$
transport
network
link
physical
client$_2$
application
transport
network
link
physical

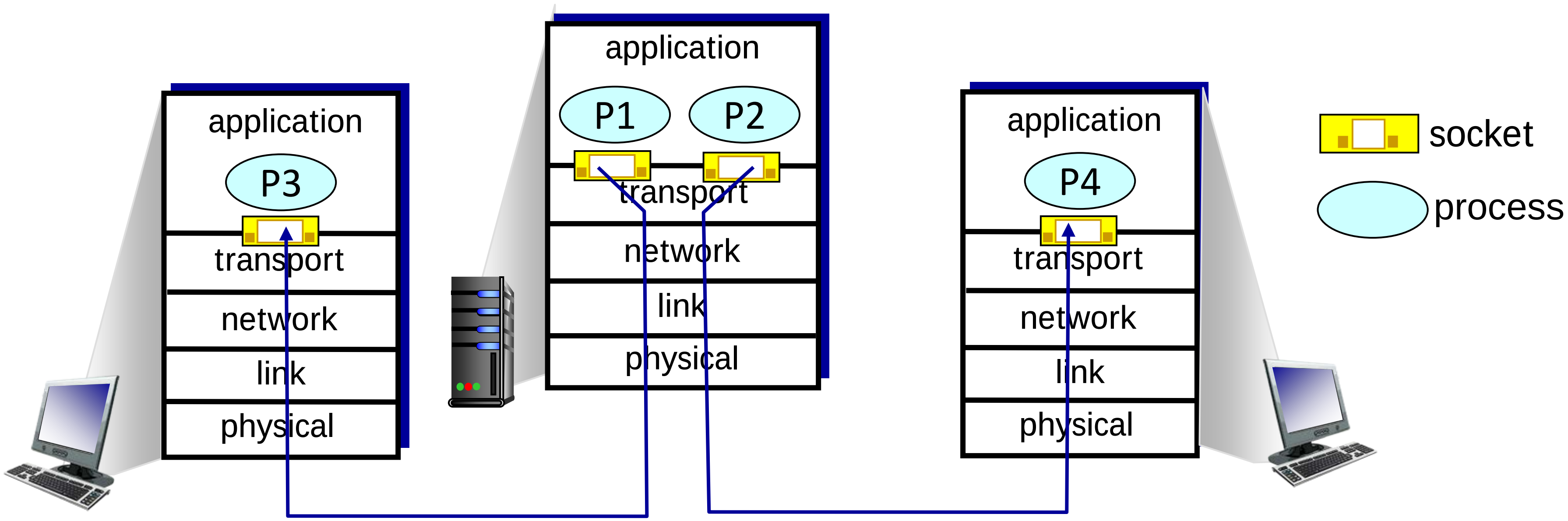
Multiplexing: cars coming from multiple ramps onto the main highway

De-multiplexing: cars being directed off of a highway onto different exit ramps

De-multiplexing: being split, being directed to a certain class of service



Multiplexing/demultiplexing



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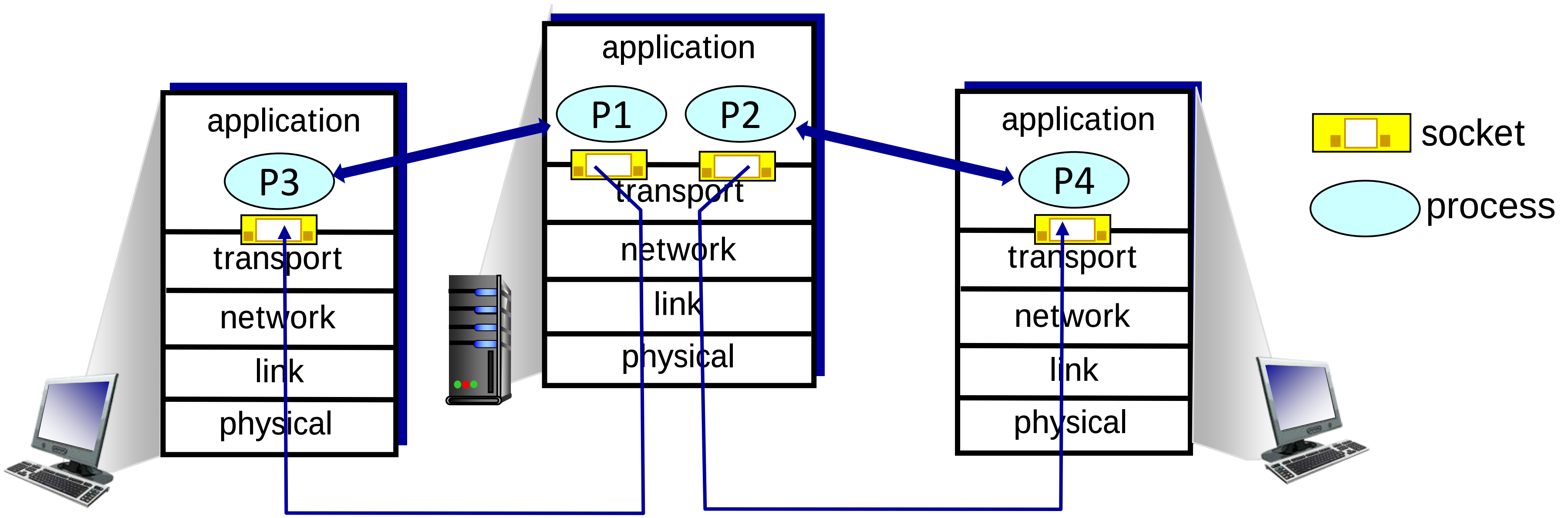
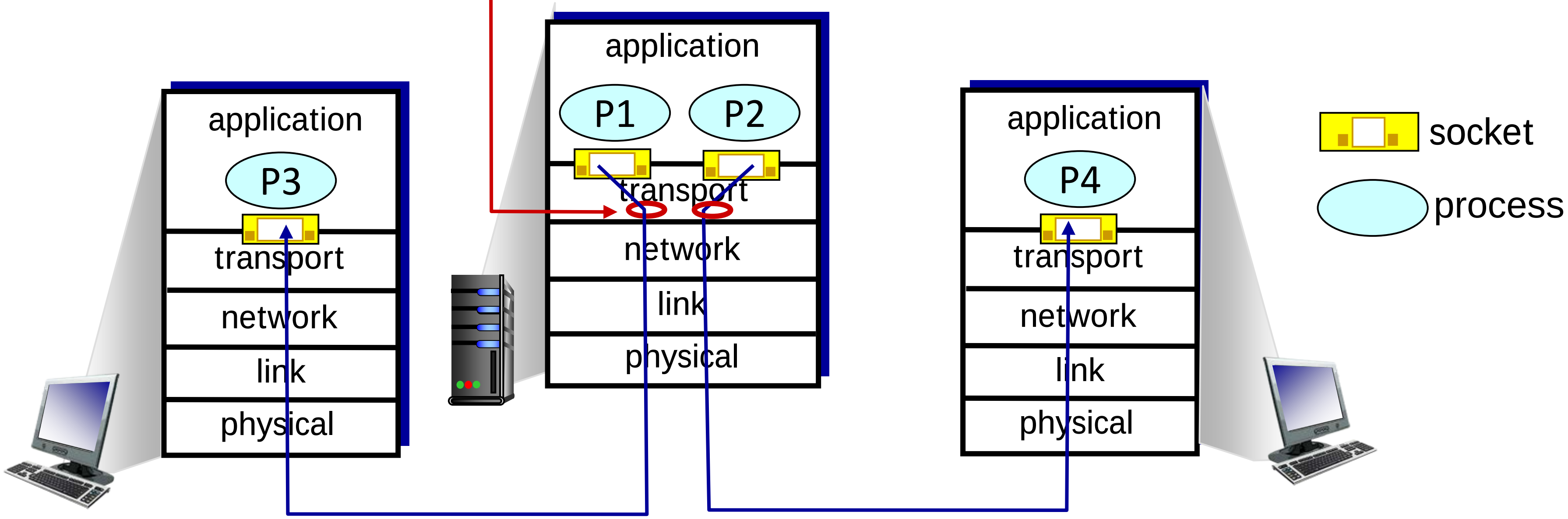


Multiplexing/demultiplexing

multiplexing as sender:

handle data from multiple sockets, add transport header (later used for demultiplexing)

P1 and P2 send through their sockets

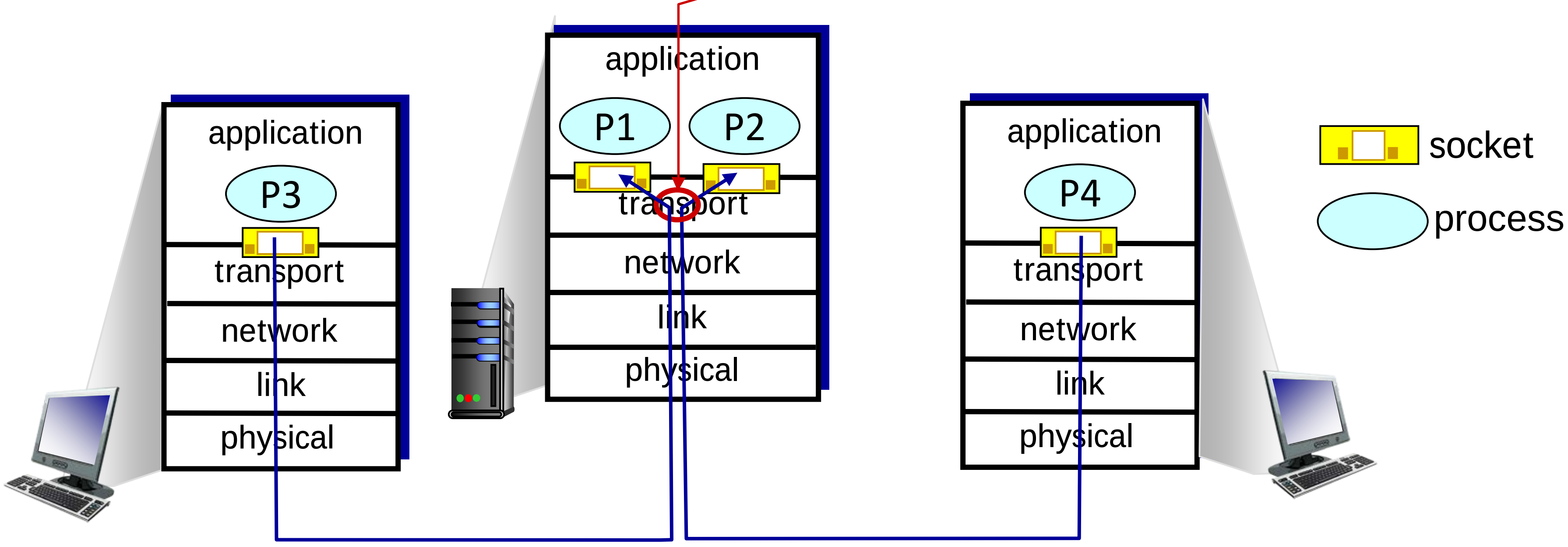


Multiplexing/demultiplexing

demultiplexing as receiver:

use header info to deliver received segments to correct socket

P1 and P2 receive through their sockets



How demultiplexing works

- host receives IP datagrams
  - each datagram has source IP address, destination IP address
  - each datagram carries one transport-layer segment
  - each segment has source, destination port number
- host uses *IP addresses & port numbers* to direct segment to appropriate socket

How demultiplexing works

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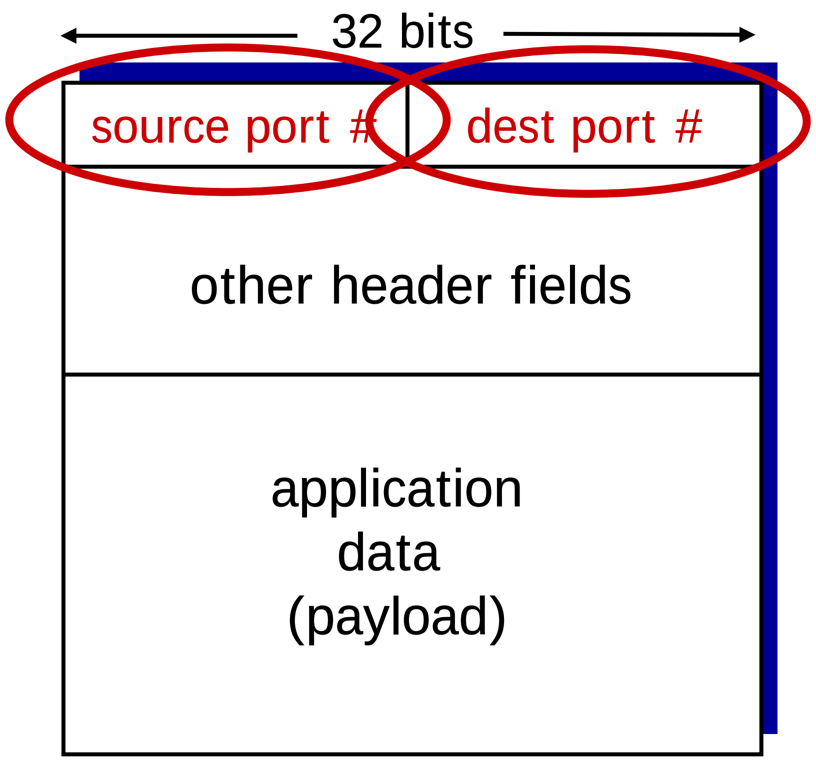

TCP/UDP segment format

Connectionless demultiplexing

Recall:

- when creating socket, must specify *host-local* port #:

```
DatagramSocket mySocket1
  = new DatagramSocket(12534);
```

-data gram socket is created,

-host local port number is specified

Connectionless demultiplexing

Recall:

- when creating socket, must specify *host-local* port #:

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- when creating datagram to send into UDP socket, must specify
  - destination IP address
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when receiving host receives *UDP* segment:

- checks destination port # in segment
- directs UDP segment to socket with that port #

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when receiving host receives *UDP* segment:

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- directs UDP segment to socket with that port #

IP/UDP datagrams with *same dest. port #,* but different source IP addresses and/or source port numbers will be directed to *same socket* at receiving host

Connectionless demultiplexing: an example

```
mySocket =
 socket(AF_INET,SOCK_DGRAM)
mySocket.bind(myaddr,6428);
```

```
mySocket =
 socket(AF_INET,SOCK_STREAM)
mySocket.bind(myaddr,9157);
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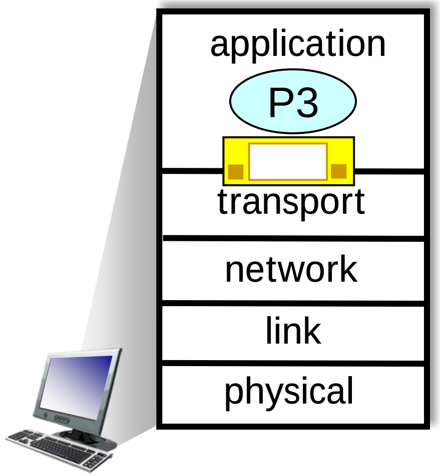

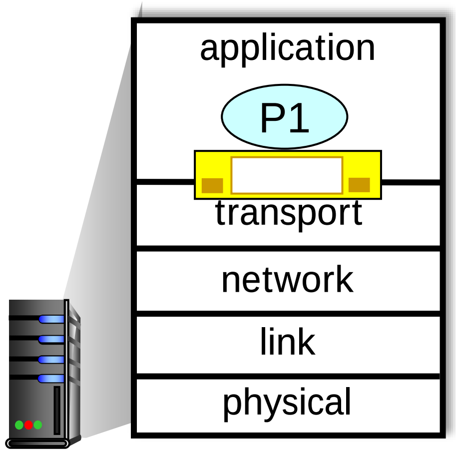

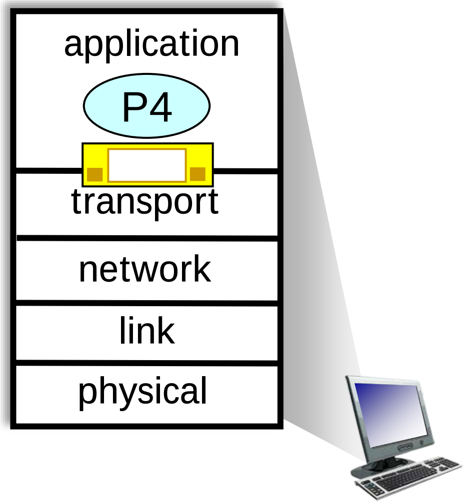

Communication:

P1 and P3

P1 and P4

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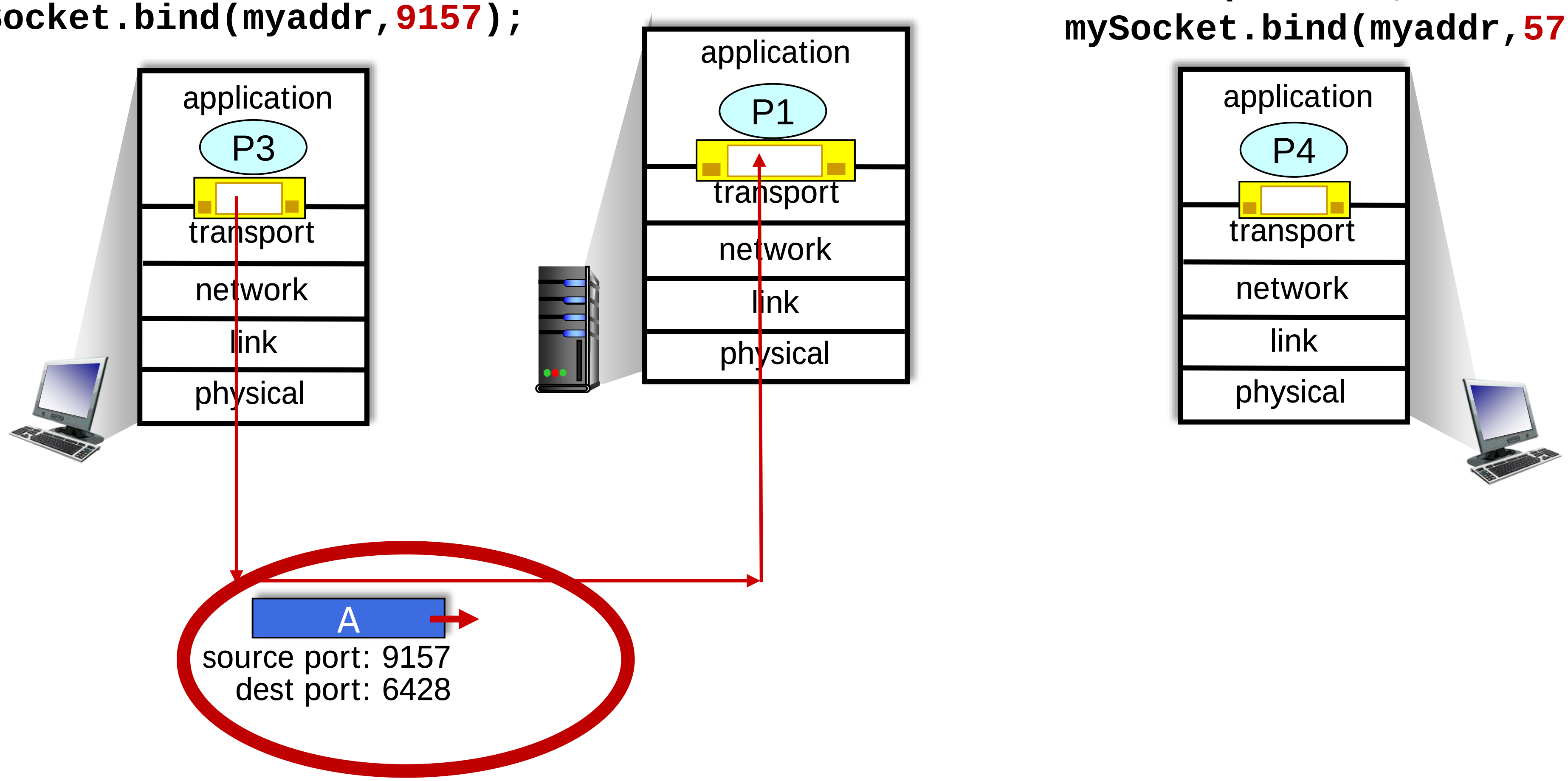

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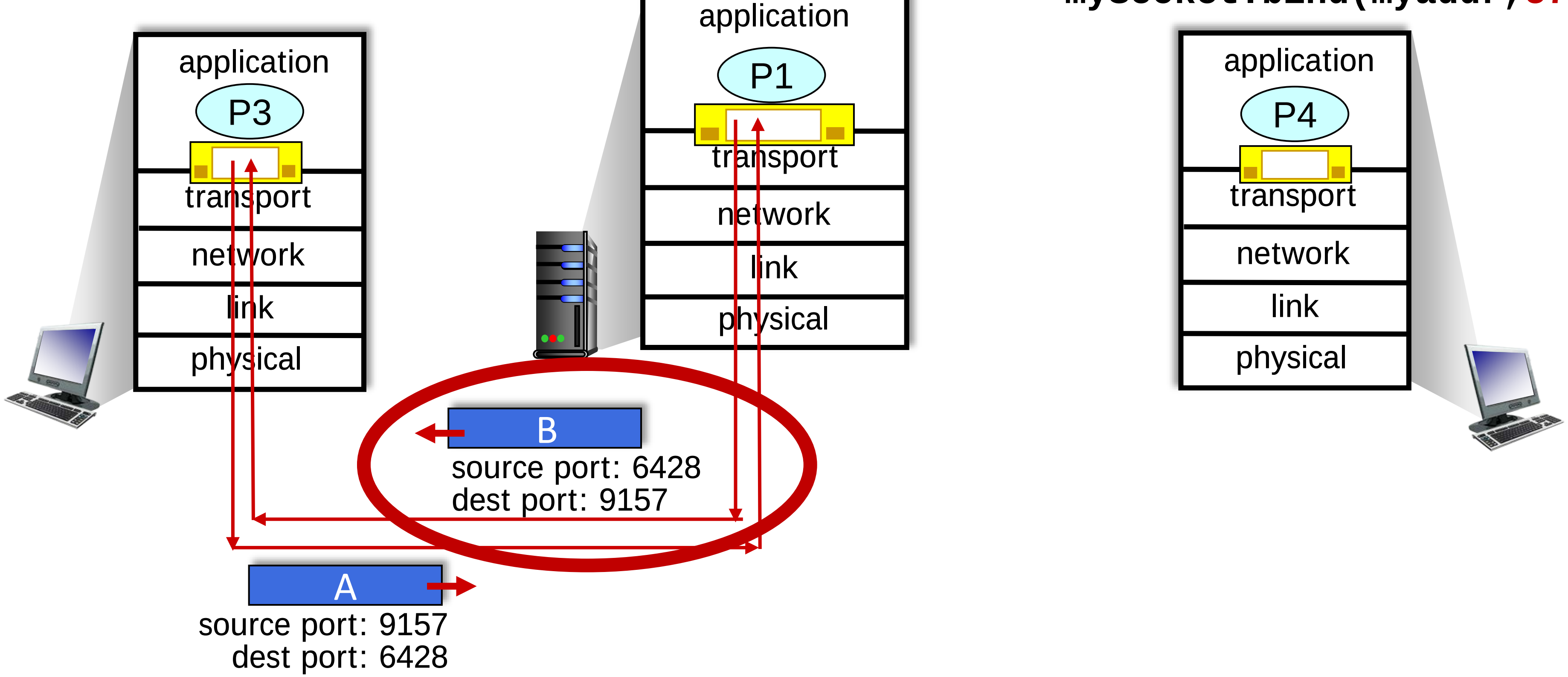

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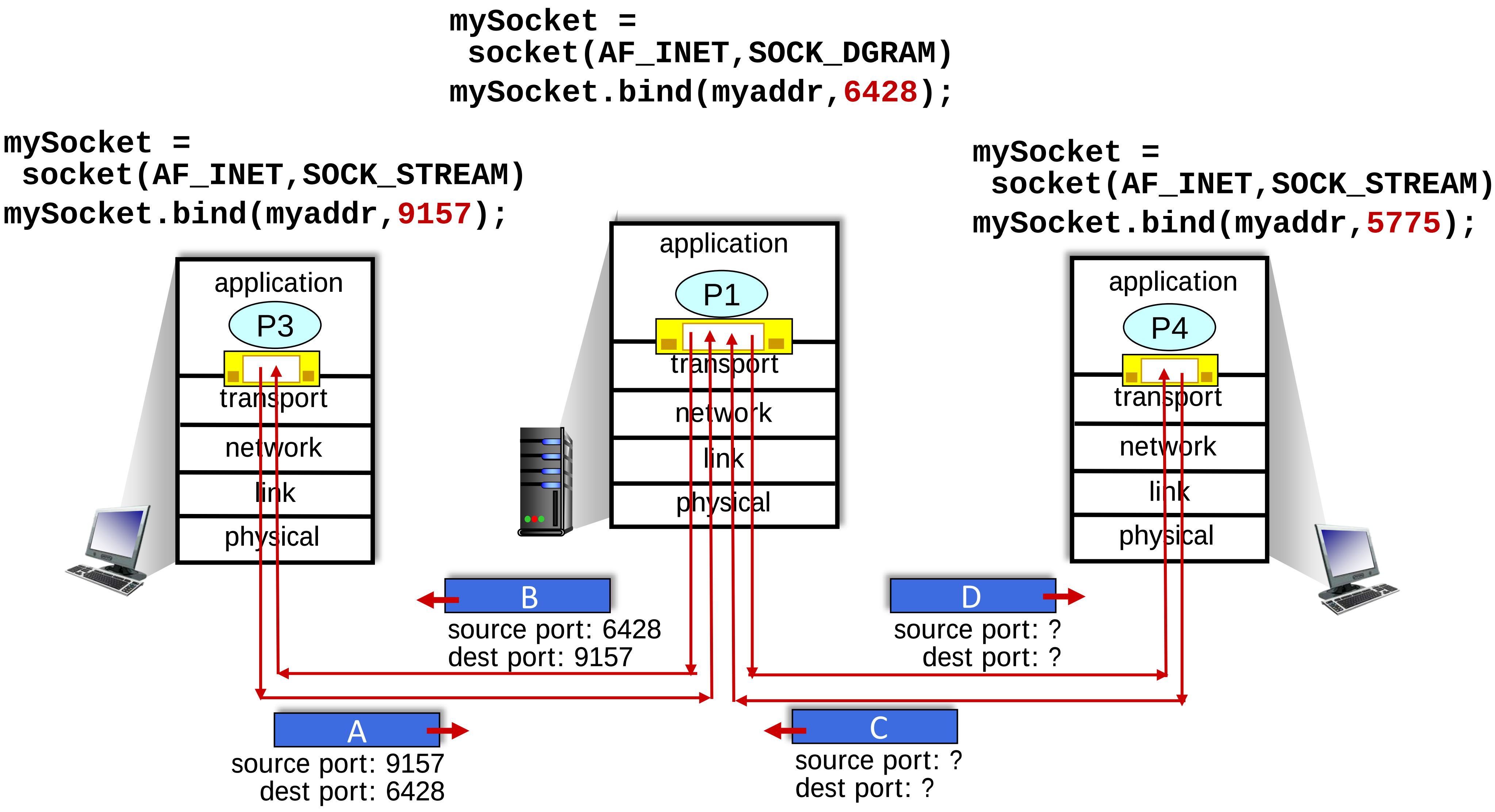

Connection-oriented demultiplexing

- TCP socket identified by 4-tuple:
  - source IP address
  - source port number
  - dest IP address
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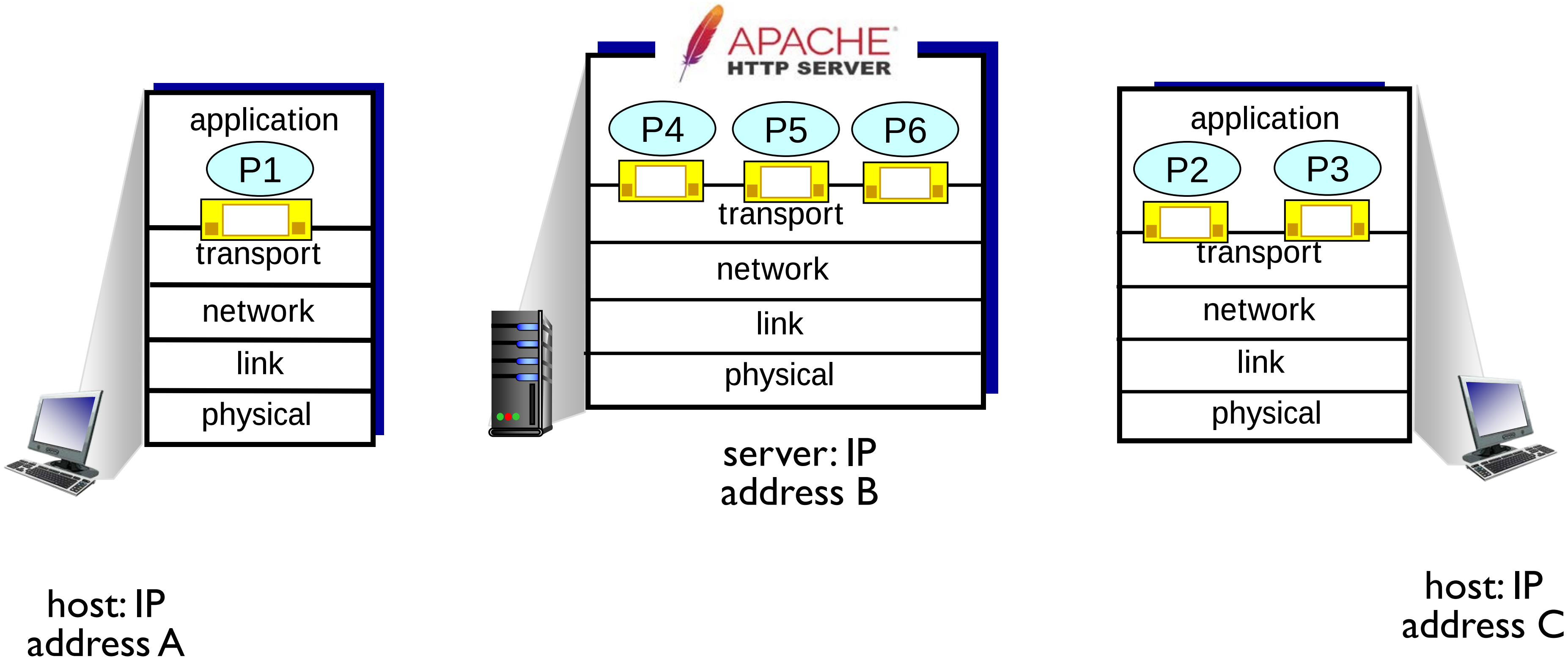
- demux: receiver uses *all four values (4-tuple)* to direct segment to appropriate socket

Connection-oriented demultiplexing

- TCP socket identified by 4-tuple:
  - source IP address
  - source port number
  - dest IP address
  - dest port number
- demultiplexing: receiver uses *all four values (4-tuple)* to direct segment to appropriate socket

- server may support many simultaneous TCP sockets:
  - each socket identified by its own 4-tuple
  - each socket associated with a different connecting client

Connection-oriented demultiplexing: example



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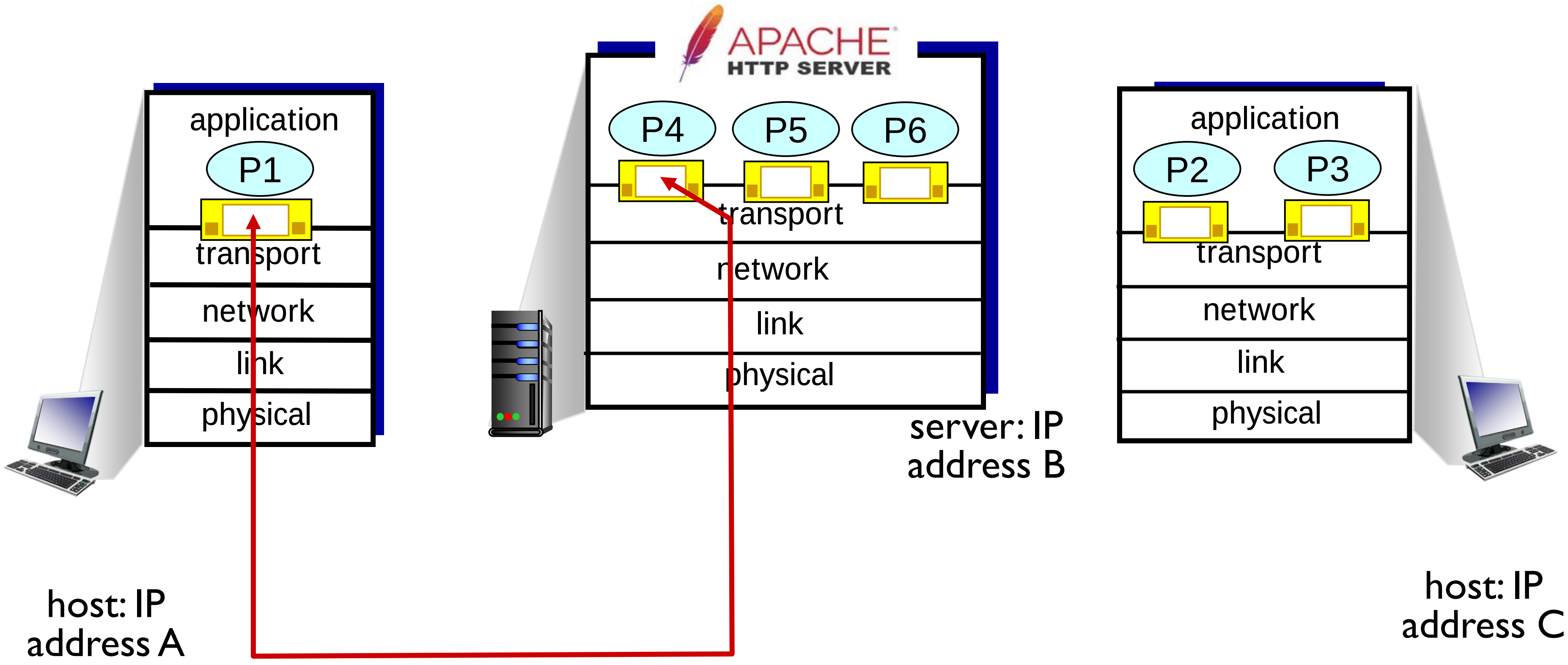

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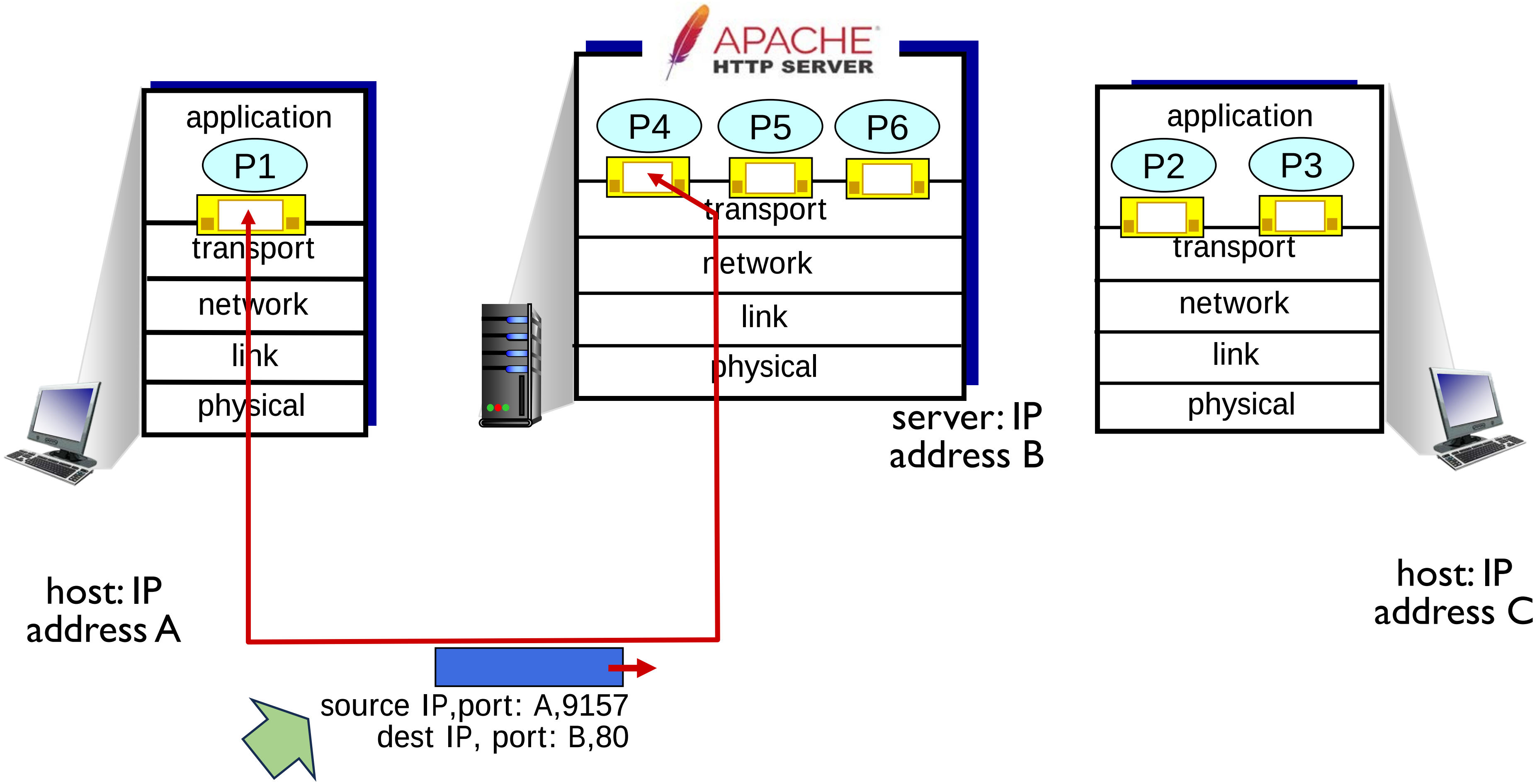

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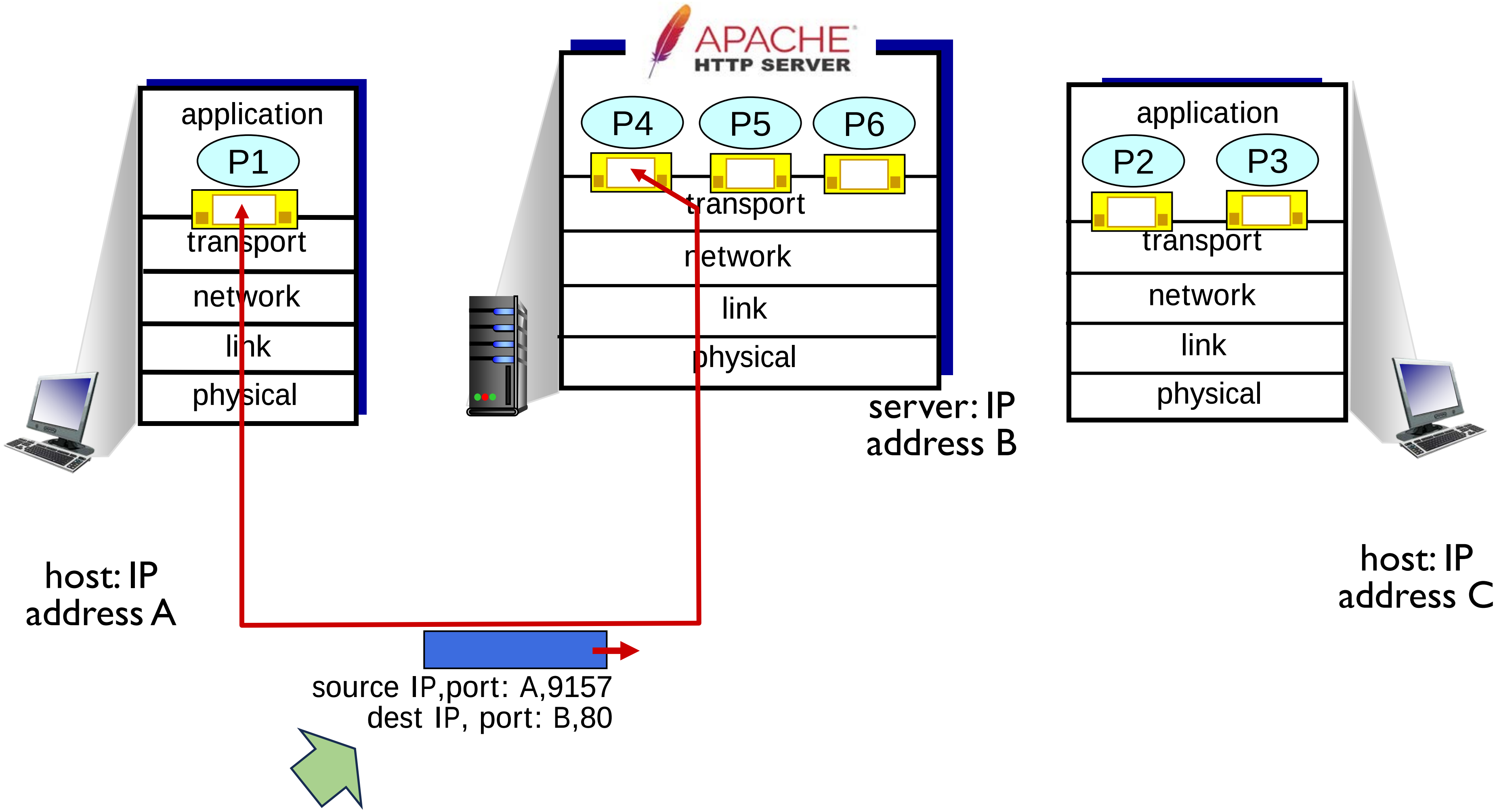

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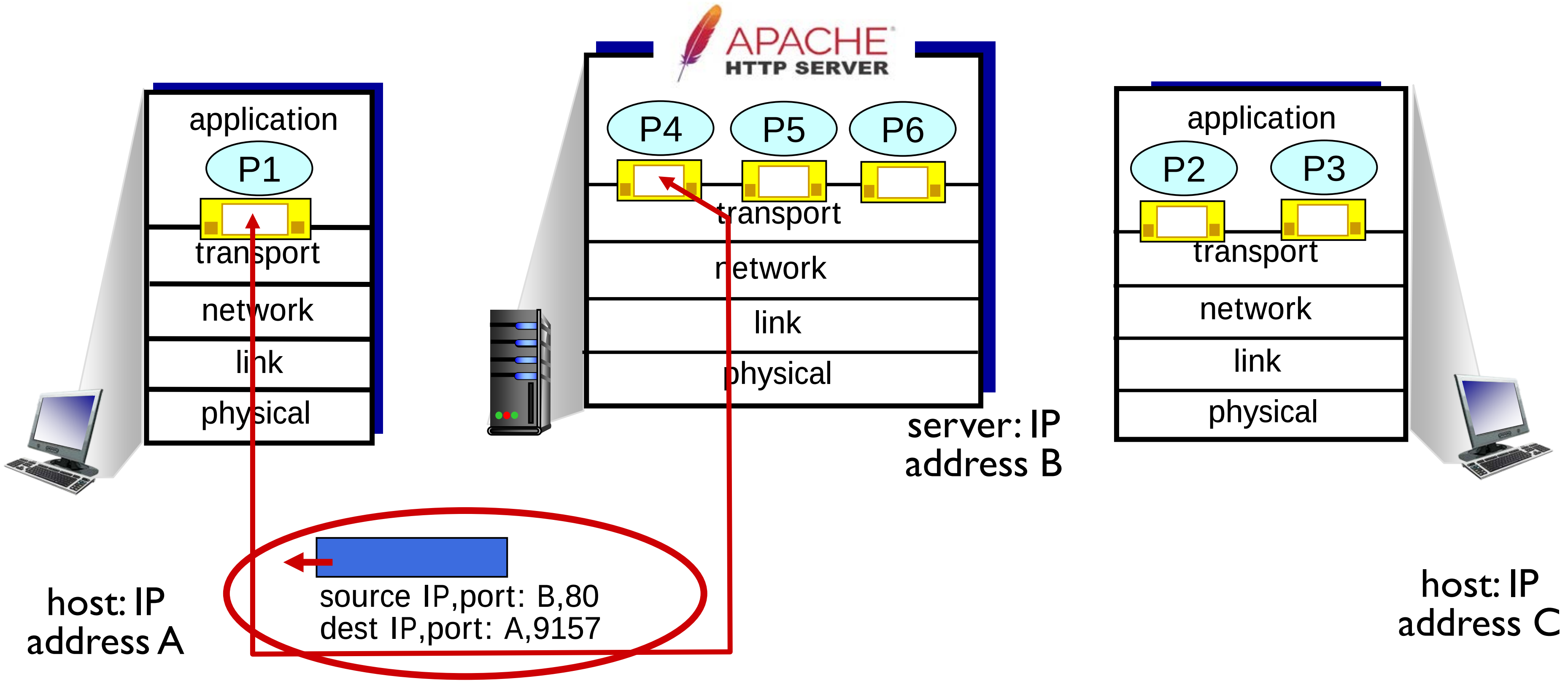

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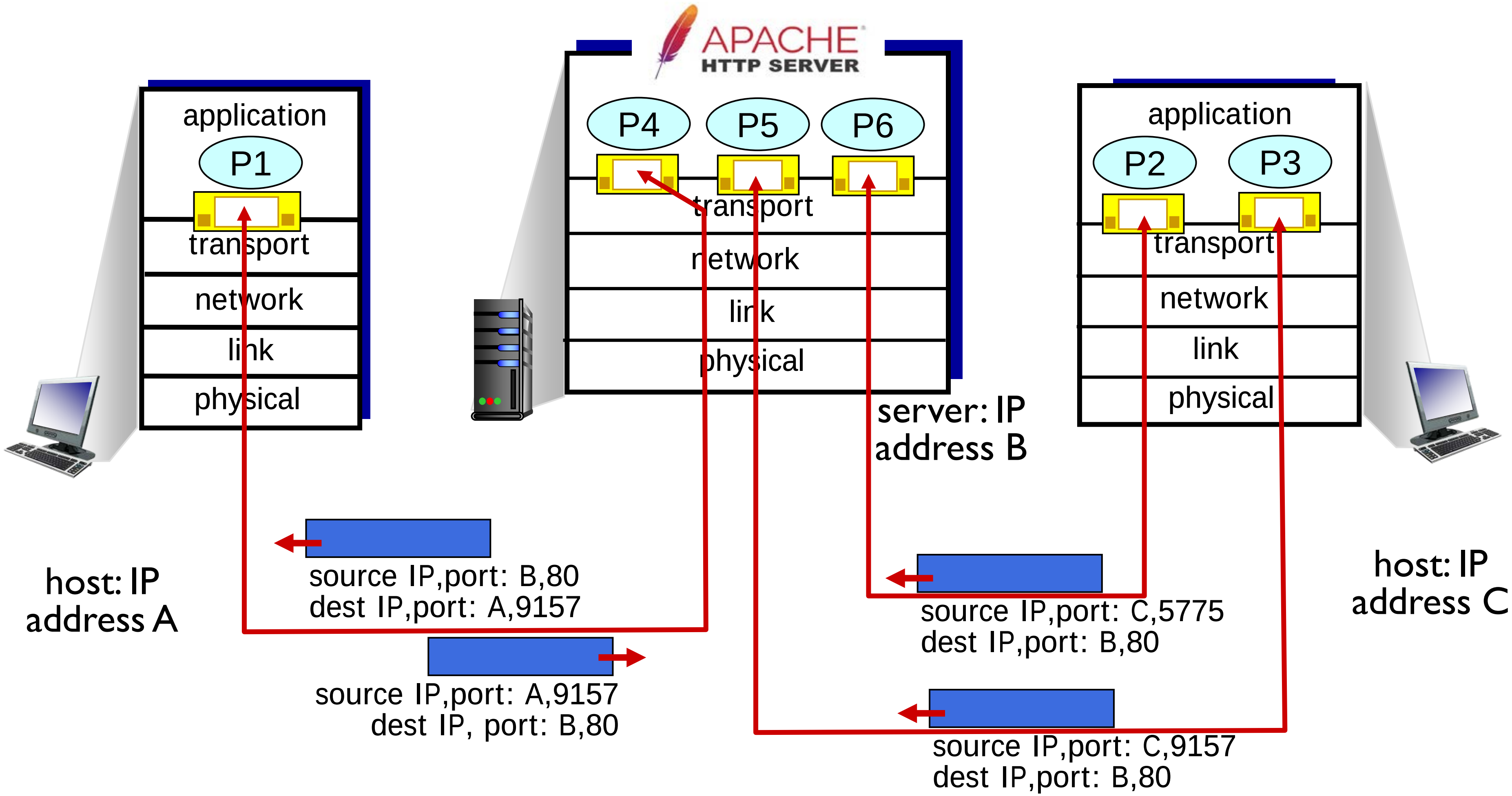
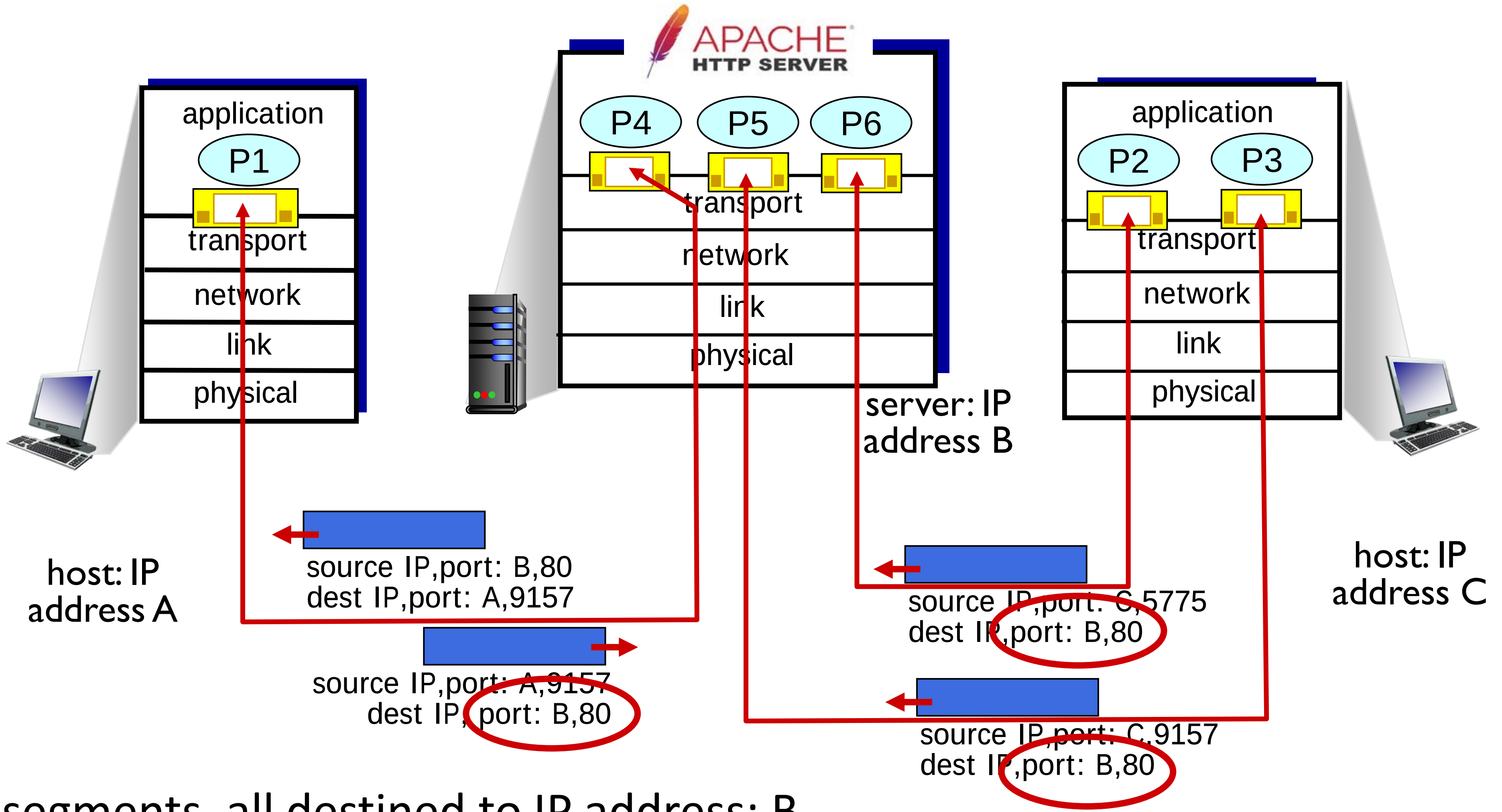


Connection-oriented demultiplexing: example



Three segments, all destined to IP address: B,
dest port: 80 are demultiplexed to *different* sockets

Demultiplexing UDP vs TCP

- UDP
  - Connectionless demultiplexing
  - Demultiplexing on the basis of the destination port ID in the datagram
- TCP
  - Connection oriented demultiplexing
  - Demultiplexing on the basis of the TCP socket four tuple
    - source IP address
    - source port number
    - dest IP address
    - dest port number

Summary

- Multiplexing, demultiplexing: based on segment, datagram header field values
- **UDP:** demultiplexing using destination port number (only)
- **TCP:** demultiplexing using 4-tuple: source and destination IP addresses, and port numbers
- Multiplexing/demultiplexing happen at *all* layers

Chapter 3: roadmap

- Transport-layer services
- Multiplexing and demultiplexing
- Connectionless transport: UDP
- Principles of reliable data transfer
- Connection-oriented transport: TCP
- Principles of congestion control
- TCP congestion control
- Evolution of transport-layer functionality

UDP: User Datagram Protocol

- UDP sender/receiver actions
- UDP segment format
- Internet checksum

UDP: User Datagram Protocol

- “no frills,” “bare bones” Internet transport protocol
- “best effort” service, UDP segments may be:
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Why is there a UDP?

- no connection establishment (which can add RTT delay)
- simple: no connection state at sender, receiver
- small header size
- no congestion control
  - UDP can blast away as fast as desired!
  - can function in the face of congestion

UDP: User Datagram Protocol

- UDP use:
  - streaming multimedia apps (loss tolerant, rate sensitive)
  - DNS
  - SNMP
  - HTTP/3
- if reliable transfer needed over UDP (e.g., HTTP/3):
  - add needed reliability at application layer
  - add congestion control at application layer

UDP: User Datagram Protocol [RFC 768]

INTERNET STANDARD

RFC 768 J. Postel
ISI
28 August 1980

User Datagram Protocol

Introduction

This User Datagram Protocol (UDP) is defined to make available a datagram mode of packet-switched computer communication in the environment of an interconnected set of computer networks. This protocol assumes that the Internet Protocol (IP) [1] is used as the underlying protocol.

This protocol provides a procedure for application programs to send messages to other programs with a minimum of protocol mechanism. The protocol is transaction oriented, and delivery and duplicate protection are not guaranteed. Applications requiring ordered reliable delivery of streams of data should use the Transmission Control Protocol (TCP) [2].

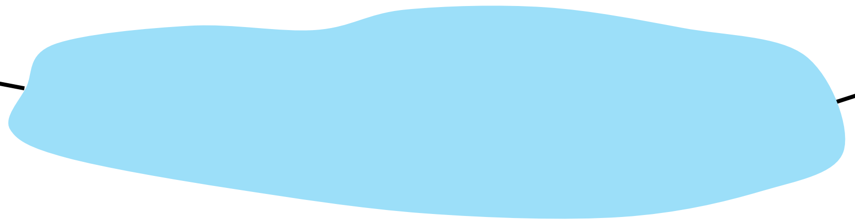
Format

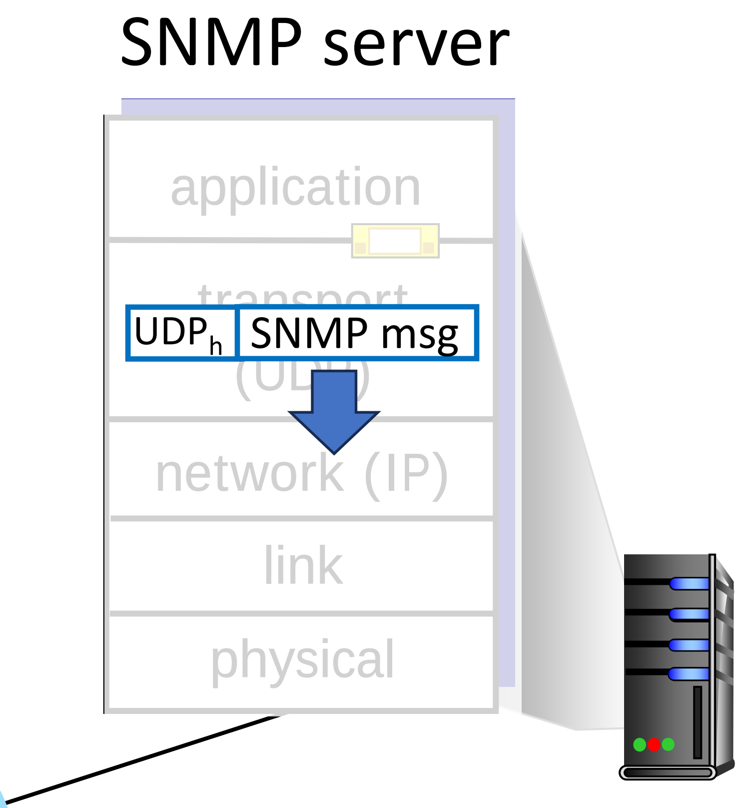
```
 0      7 8     15 16    23 24    31
+--------+--------+--------+--------+
|     Source      |   Destination   |
|      Port       |      Port       |
+--------+--------+--------+--------+
|                 |                 |
|     Length      |    Checksum     |
+--------+--------+--------+--------+
|
|          data octets ...
+---------------- ...
```

UDP: Transport Layer Actions

UDP sender actions:

- is passed an application-layer message
- determines UDP segment header fields values
- creates UDP segment
- passes segment to IP





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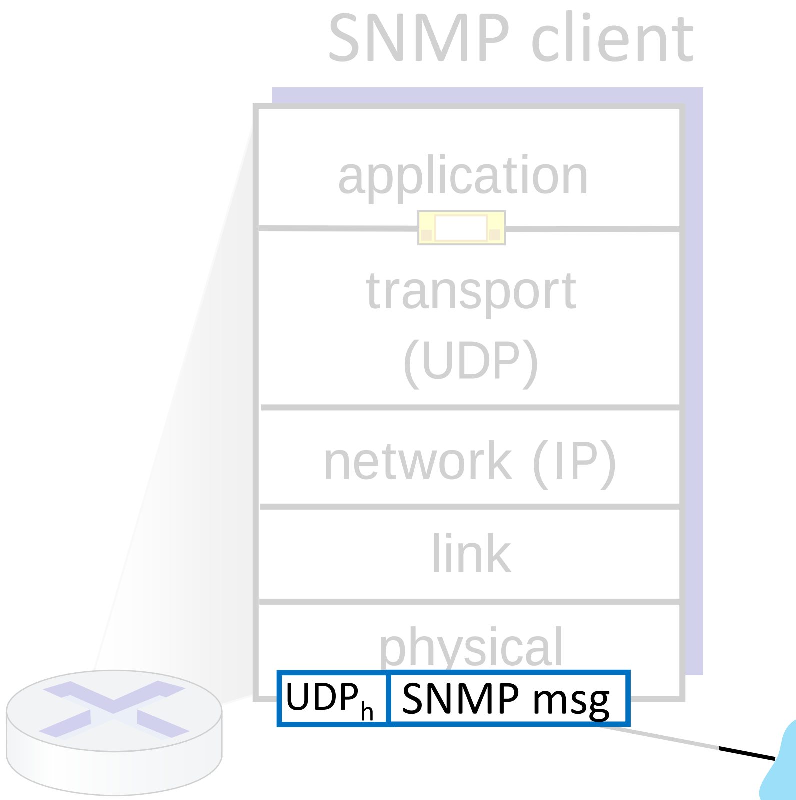

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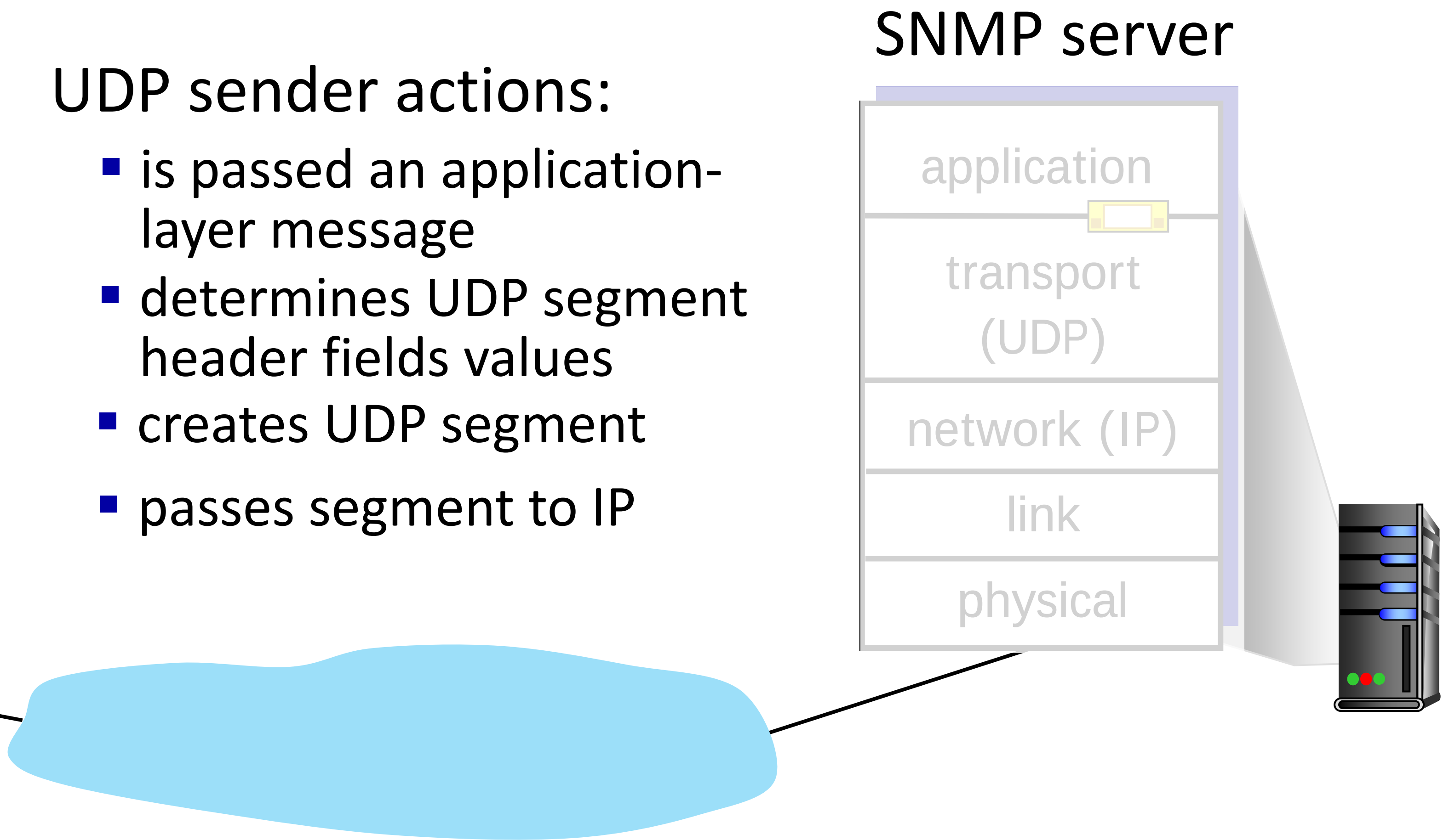

UDP: Transport Layer Actions

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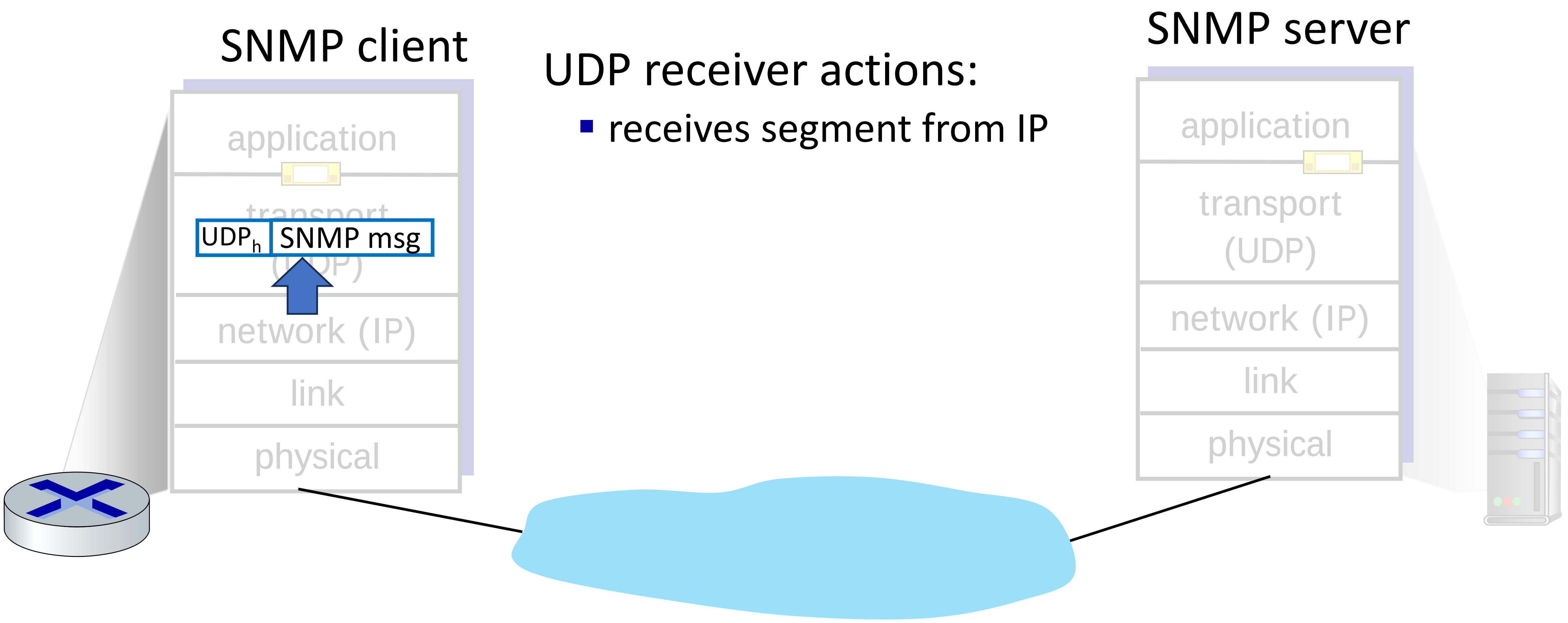

UDP: Transport Layer Actions

UDP receiver actions:

- receives segment from IP
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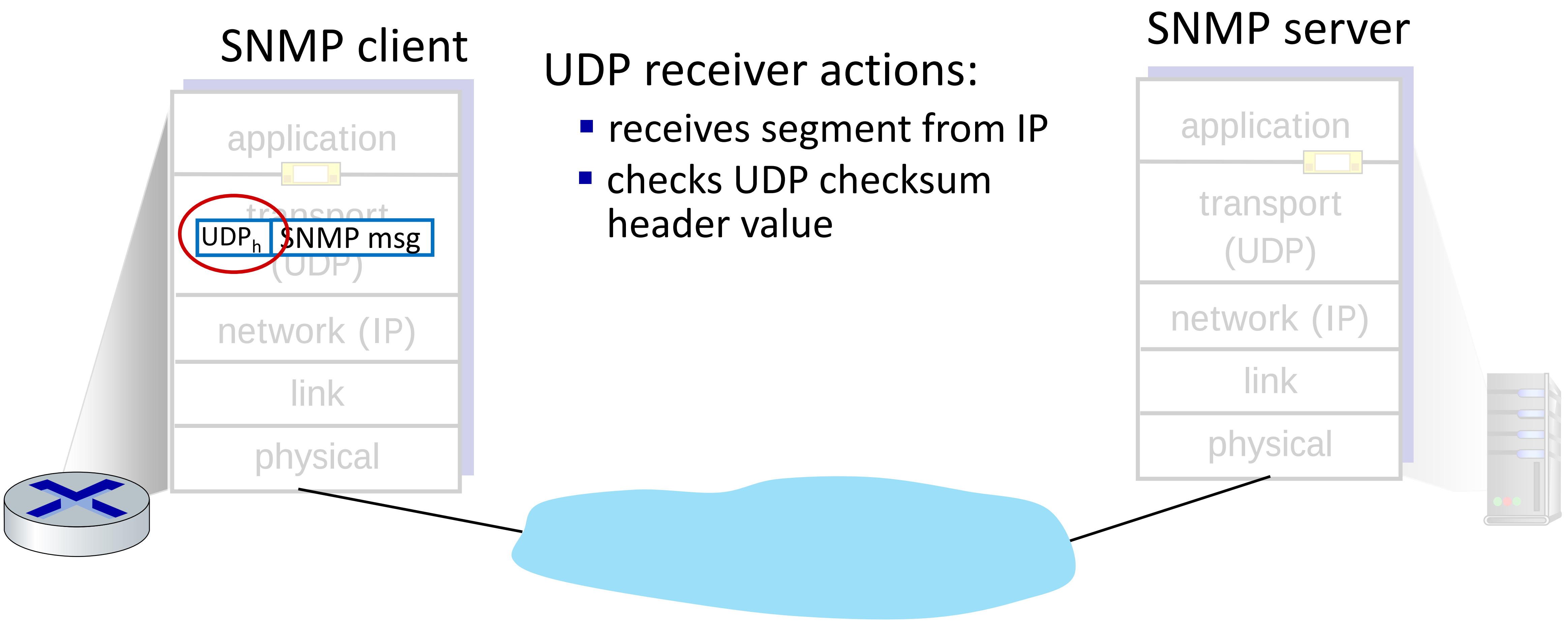

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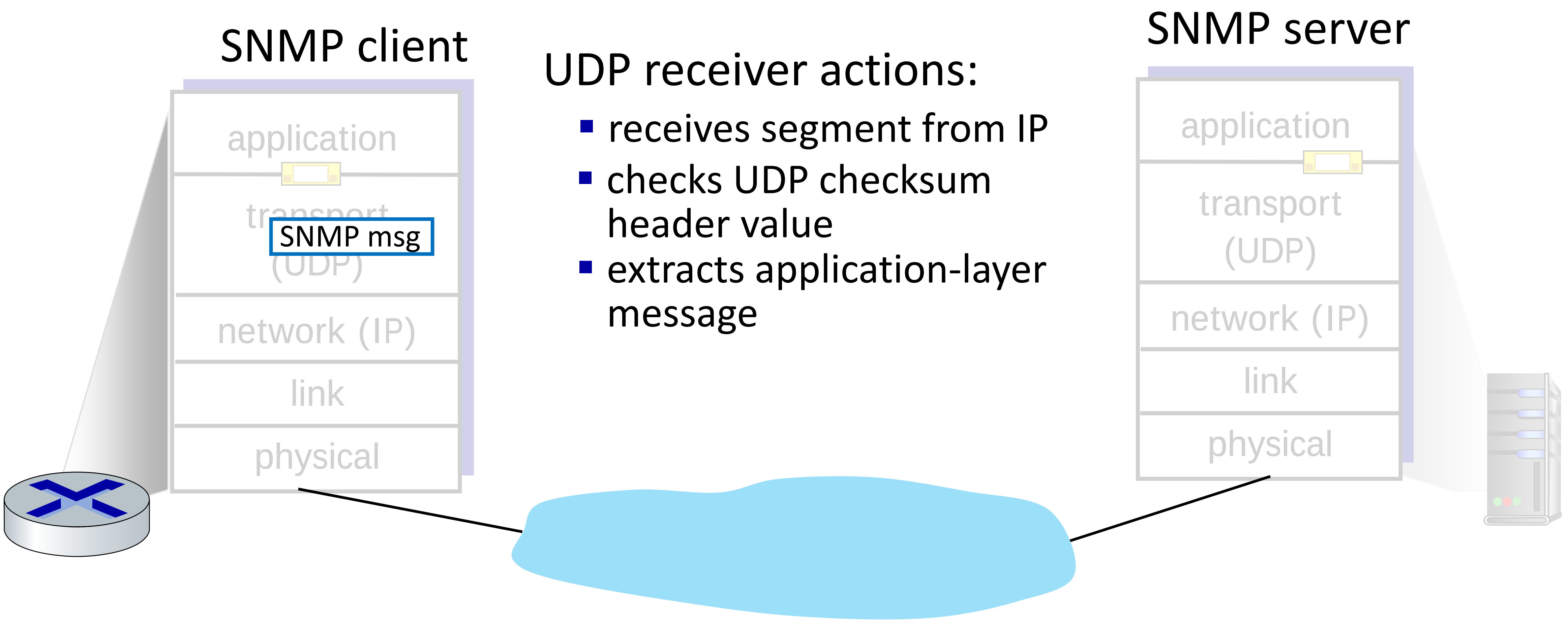

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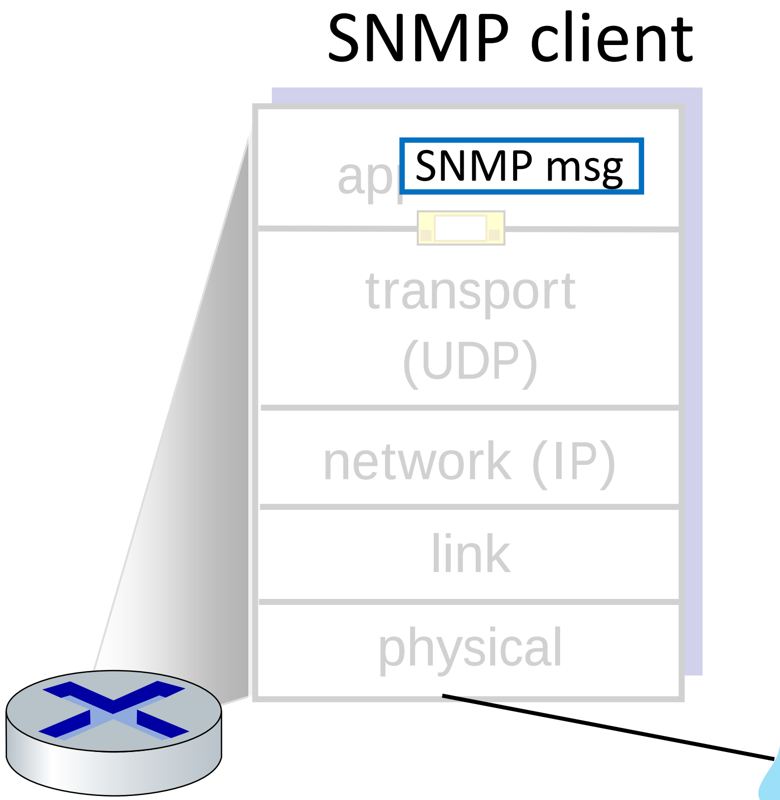
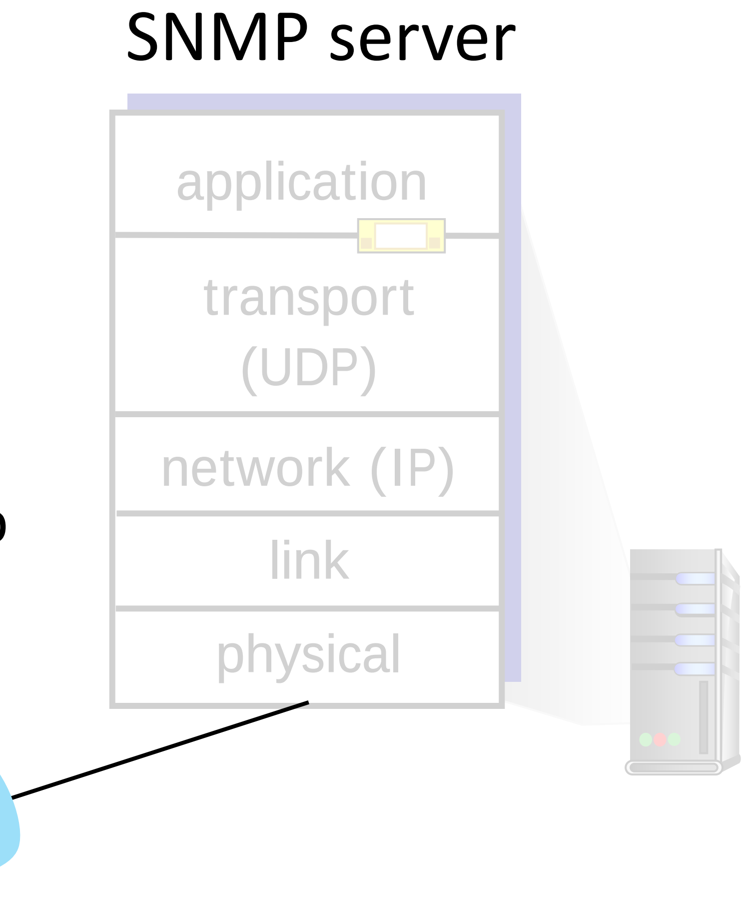
UDP: Transport Layer Actions

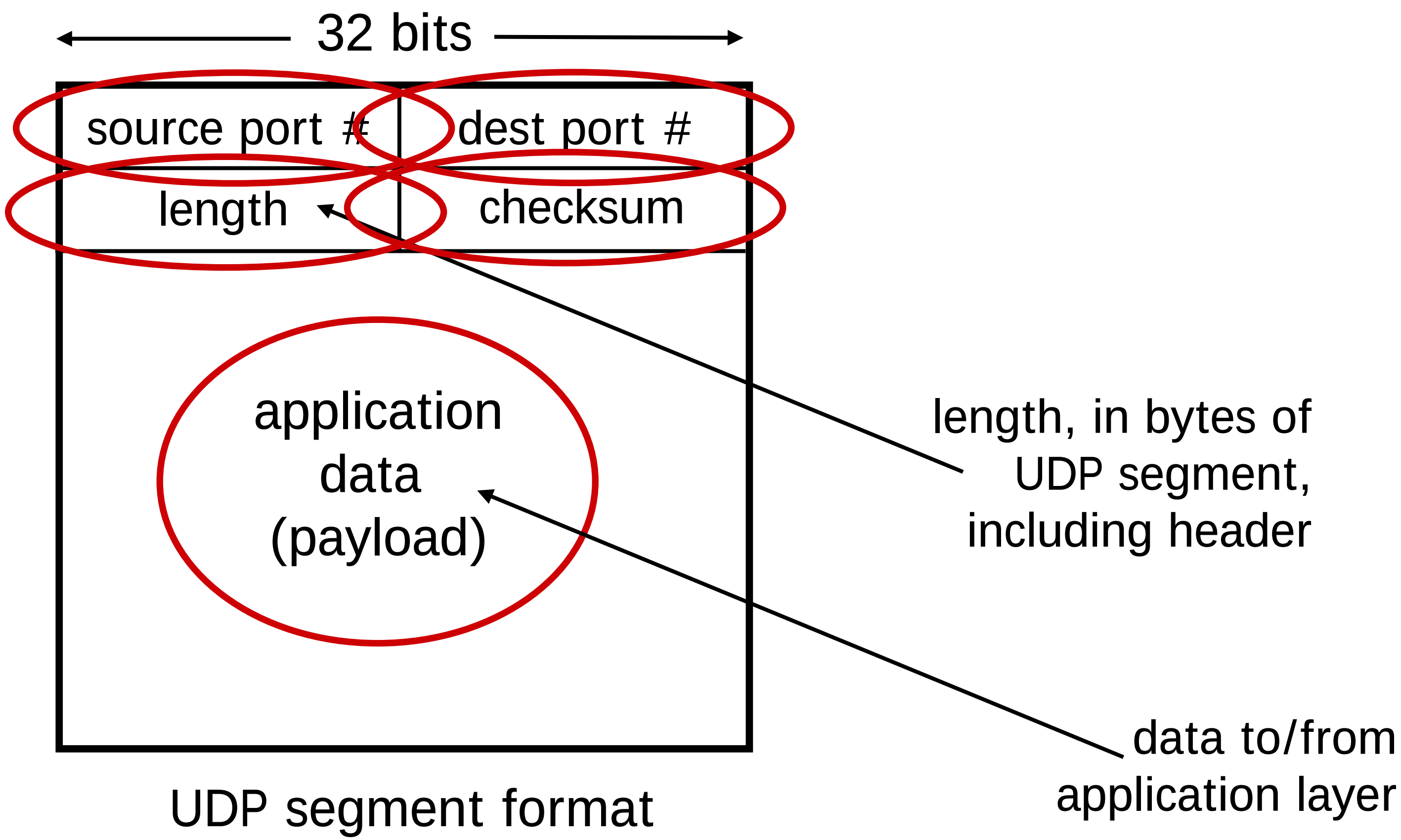


UDP receiver actions:

- receives segment from IP
- checks UDP checksum header value
- extracts application-layer message
- demultiplexes message up to application via socket



UDP segment header



UDP segment format

UDP checksum

Goal: detect errors (*i.e.*, flipped bits) in transmitted segment

UDP checksum

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	1st number	2nd number	sum
Transmitted:	5	6	11
Received:	4	6	11

receiver-computed checksum ≠ sender-computed checksum (as received)

Internet checksum

Goal: detect errors (*i.e.,* flipped bits) in transmitted segment

sender:

- treat contents of UDP segment (including UDP header fields and IP addresses) as sequence of 16-bit integers
- checksum: addition (one's complement sum) of segment content
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receiver:

- compute checksum of received segment
- check if computed checksum equals checksum field value:
  - not equal - error detected
  - equal - no error detected. *But maybe errors nonetheless?* More later ….

Internet checksum: an example

example: add two 16-bit integers

```
  1 1 1 0 0 1 1 0 0 1 1 0 0 1 1 0
  1 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
----------------------------------
1 1 0 1 1 1 0 1 1 1 0 1 1 1 0 1 1
```

Add the integers together using base two arithmetic

* Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/interactive/

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Internet checksum: an example

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		1	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0
		1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1
wraparound	1	1	0	1	1	1	0	1	1	1	0	1	1	1	0	1	1
sum		1	0	1	1	1	0	1	1	1	0	1	1	1	1	0	0
checksum		0	1	0	0	0	1	0	0	0	1	0	0	0	0	1	1

Note: when adding numbers, a carryout from the most significant bit needs to be added to the result

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Internet checksum: weak protection!

example: add two 16-bit integers

```
                1 1 1 0 0 1 1 0 0 1 1 0 0 1 1 0     0 1
                1 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1     1 0
              -----------------------------------
wraparound    1 1 0 1 1 1 0 1 1 1 0 1 1 1 0 1 1
              -----------------------------------
sum             1 0 1 1 1 0 1 1 1 0 1 1 1 1 0 0
checksum        0 1 0 0 0 1 0 0 0 1 0 0 0 0 1 1
```

Internet checksum: weak protection!

example: add two 16-bit integers

	1 1 1 0 0 1 1 0 0 1 1 0 0 1 1 0	0 1
	1 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1	1 0
wraparound	1 1 0 1 1 1 0 1 1 1 0 1 1 1 0 1 1	
sum	1 0 1 1 1 0 1 1 1 0 1 1 1 1 0 0	
checksum	0 1 0 0 0 1 0 0 0 1 0 0 0 0 1 1	

Even though numbers have changed (bit flips), *no* change in checksum!

Internet checksum not fully foolproof.

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  - helps with reliability (checksum)
- build additional functionality on top of UDP in application layer (e.g., HTTP/3)

That's all for now

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